Qualcomm Technologies, Inc.

IPQ8078 Wi-Fi Access Point SoC

Device Specification

80-YA726-4 Rev. C

January 29, 2018

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

For additional information or to submit technical questions go to https://createpoint.qti.qualcomm.com

Confidential and Proprietary – Qualcomm Technologies, Inc.

NO PUBLIC DISCLOSURE PERMITTED: Please report postings of this document on public servers or websites to: DocCtrlAgent@qualcomm.com.

Restricted Distribution: Not to be distributed to anyone who is not an employee of either Qualcomm Technologies, Inc. or its affiliated companies without the express approval of Qualcomm Configuration Management.

Not to be used, copied, reproduced, or modified in whole or in part, nor its contents revealed in any manner to others without the express written permission of Qualcomm Technologies, Inc.

Qualcomm is a trademark of Qualcomm Incorporated, registered in the United States and other countries. Other product and brand names may be trademarks or registered trademarks of their respective owners.

This technical data may be subject to U.S. and international export, re-export, or transfer (“export”) laws. Diversion contrary to U.S. and international law is strictly prohibited.

Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

© 2017-2018 Qualcomm Technologies, Inc. All rights reserved.

Revision history

Revision	Date	Description
A	July 2017	Initial release
B	November 2017	■ Mechanical information: Updated PRR code in Table 4-2 and Table 4-4.
C	January 2018	■ Introduction: Updated A53 frequency to 2.2 GHz, NPU frequency to 1.7 GHz, and DDR4 rate to 2400 MT/s.

QUALCOMM
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

Contents

1 Introduction 8
- 1.1 Functional block diagram 8
- 1.2 Wi-Fi subsystem 8
- 1.3 Networking subsystem 9
- 1.4 CPU subsystem 9
- 1.5 Interfaces and power management 9
- 1.6 IPQ8078 features 10
  - 1.6.1 Wi-Fi subsystem 10
  - 1.6.2 Networking subsystem 10
  - 1.6.3 CPU subsystem 11
  - 1.6.4 Peripherals/interfaces 12
  - 1.6.5 Power management 12
  - 1.6.6 Platform extension options 12
  - 1.6.7 Package 12
- 1.7 Terms and abbreviations 12

2 Pin definitions 14
- 2.1 I/O parameter definitions 14
- 2.2 Pin map 15
- 2.3 Pin descriptions 17
  - 2.3.1 CLK/RST and PMIC interface 17
  - 2.3.2 PCI express endpoint 17
  - 2.3.3 Wi-Fi 5 GHz PHY 18
  - 2.3.4 Wi-Fi 2.4 GHz PHY 20
  - 2.3.5 Analog test 21
  - 2.3.6 DDR4/DDR3L 21
  - 2.3.7 Mode 24
  - 2.3.8 PLL test clock 24
  - 2.3.9 PSGMII 25
  - 2.3.10 SDC 25
  - 2.3.11 JTAG 26
  - 2.3.12 USB 26
  - 2.3.13 USXGMII 27
  - 2.3.14 GPIO 27

QUALCOMM® 2018-02-01 09:07:08 PST steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2.3.15 Ground, power-supply and NC . . . 37
2.3.16 Boot configuration GPIOs . . . 40

3 Electrical specifications . . . 42
3.1 Absolute maximum ratings . . . 42
3.2 Operating conditions . . . 43
3.3 Power consumption . . . 45
3.4 Power sequencing . . . 45
3.5 Digital-logic characteristics . . . 47
3.6 Timing characteristics . . . 47
3.7 Memory support . . . 47
3.8 Connectivity . . . 47
3.9 UNIPHY interfaces . . . 47
3.10 Internal functions . . . 47
3.10.1 Clocks . . . 47
3.10.2 Modes and resets . . . 48
3.10.3 JTAG . . . 49
3.11 Power management interfaces . . . 49
3.11.1 SPMI . . . 49
3.12 Wi-Fi and Analog interfaces . . . 49

4 Mechanical information . . . 50
4.1 Device physical dimensions . . . 50
4.2 Part marking . . . 52
4.2.1 Specification-compliant devices . . . 52
4.3 Device ordering information . . . 53
4.4 Device moisture-sensitivity level . . . 54
4.5 Thermal characteristics . . . 54

5 Carrier, storage, and handling information . . . 55
5.1 Carrier . . . 55
5.1.1 Tape and reel information . . . 55
5.1.2 Matrix tray information . . . 56
5.2 Storage . . . 57
5.2.1 Bag storage conditions . . . 57
5.2.2 Out of bag duration . . . 57
5.3 Handling . . . 57
5.3.1 Baking . . . 57
5.3.2 Electrostatic discharge . . . 57
5.4 Barcode label and packing for shipment . . . 58

6 PCB mounting guidelines . . . 59
6.1 RoHS compliance . . . 59
6.2 SMT parameters . . . 59

QUALCOMM® 2013-02-01 09:07:08 PST Devi.umsby@spirent.com

6.2.1 Land pad and stencil design 59
6.2.2 Reflow profile 60
6.2.3 SMT peak package-body temperature 61
6.2.4 SMT process verification 61
6.2.5 Board-level reliability 61
6.3 High-temperature warpage 62

7 Part reliability 63
7.1 Reliability qualifications summary 63
7.2 Qualification sample description 63

QUALCOMM
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

Figures

Figure 1-1 IPQ8078 functional block diagram 8
Figure 2-1 IPQ8078 pin assignments 16
Figure 3-1 Power-on sequence 46
Figure 3-2 XO timing parameters 47
Figure 3-3 Sleep-clock timing parameters 48
Figure 3-4 JTAG interface timing diagram 49
Figure 4-1 IPQ8078 mechanical dimensions, top and bottom views 51
Figure 4-2 IPQ8078 device marking 52
Figure 4-3 Device identification code 53
Figure 5-1 Tape orientation on reel 55
Figure 5-2 Part orientation in tape 56
Figure 5-3 Matrix tray part orientation 56

QUALCOMM
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Tables

Table 1-1 Terms and abbreviations 13
Table 2-1 I/O description parameters 14
Table 2-2 CLK/RST and PMIC interface 17
Table 2-3 PCI express endpoint 17
Table 2-4 Wi-Fi 5 GHz PHY 18
Table 2-5 Wi-Fi 2.4 GHz PHY 20
Table 2-6 Analog test 21
Table 2-7 DDR4/DDR3L 21
Table 2-8 Mode 24
Table 2-9 PLL test clock 24
Table 2-10 PSGMII 25
Table 2-11 SDC 25
Table 2-12 JTAG 26
Table 2-13 USB 26
Table 2-14 USXMII 27
Table 2-15 GPIO pins 27
Table 2-16 Ground, power-supply and NC pins 37
Table 2-17 Boot configuration GPIOs 40
Table 3-1 Absolute maximum ratings 42
Table 3-2 Operating conditions 43
Table 3-3 Operating conditions for voltage rails with AVS 45
Table 3-4 XO timing parameters 48
Table 3-5 Sleep-clock timing parameters 48
Table 3-6 JTAG interface timing characteristics 49
Table 3-7 Supported SPMI standards and exceptions 49
Table 4-1 Package marking line description 53
Table 4-2 Device identification details 53
Table 4-3 Source configuration code 54
Table 4-4 Ordering numbers 54
Table 4-5 Device JEDEC thermal resistance 54
Table 6-1 Typical SMT reflow profile conditions (for reference only) 60

QUALCOMM®

1 Introduction

IPQ8078 is an SoC for 11ax Wi-Fi Access Points, Retail Routers and Carrier Gateways. The chip consists of a Wi-Fi subsystem, a networking subsystem and a CPU subsystem.

1.1 Functional block diagram

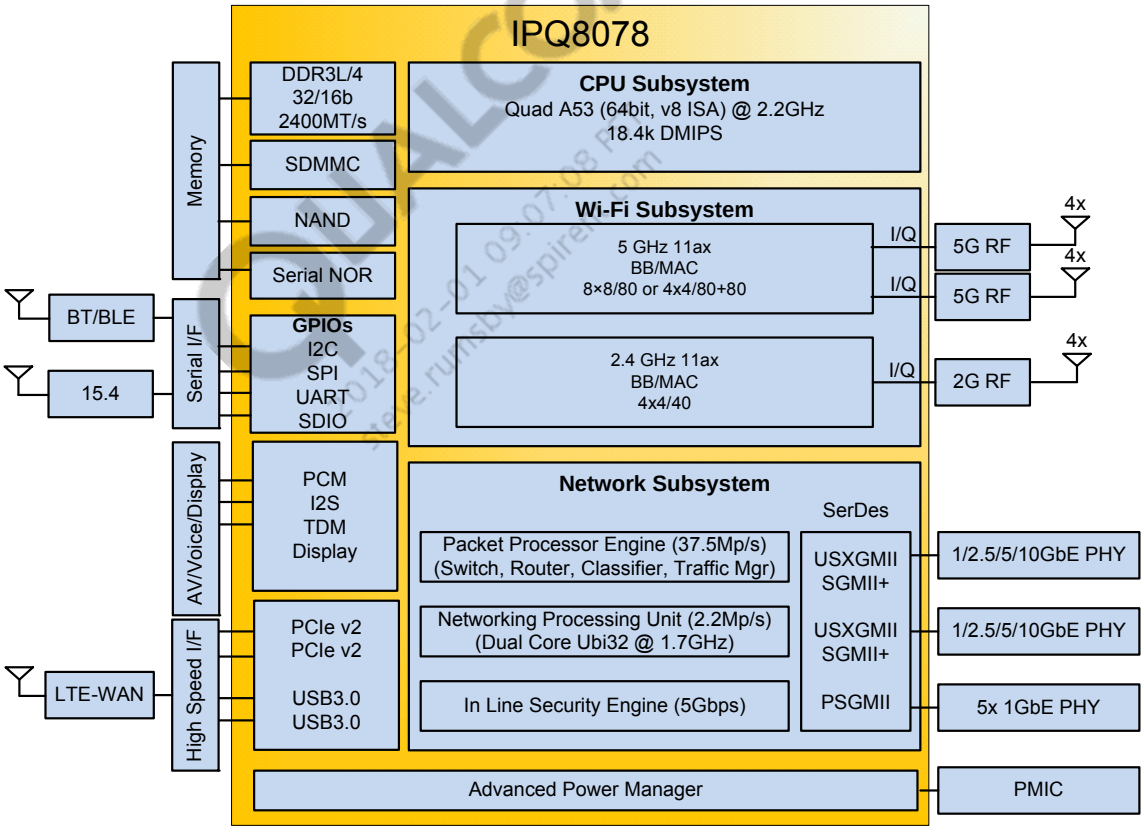


Figure 1-1 IPQ8078 functional block diagram

1.2 Wi-Fi subsystem

The Wi-Fi subsystem supports IEEE802.11ax. The IPQ8078 supports dual band dual current (DBDC) operation. 12 antenna chains operate in a two radio configuration as 8x8 in 5 GHz and 4x4 in 2.4 GHz.

The Wi-Fi PHY rates equate to 5950 Mbps, 4800 Mbps for 5 GHz and 1150 Mbps for 2.4 GHz, enabling an AX6000 product.

1.3 Networking subsystem

The networking subsystem is a high performance high throughput programmable offload engine to the networking stack that runs on the Host CPU subsystem. It interfaces to 3 Ethernet SerDes to connect to external multi-GbE PHYs. Two of the three SerDes support up to 10GbE PHY (either USXGMII, XFI, SGMII or SGMII+) while the third SerDes runs in either PSGMII, QSGMII or SGMII mode to connect to QCA8075 (5 port GbE PHY array) or QCA803x (single port GbE PHY).

The networking subsystem is capable of classifying incoming packets at an aggregate throughput rate of 25 Gbps, 37.5 million packets per second (Mpps) for 64Byte packets. This high performance ingress packet engine makes IPQ8078 very well suited to deliver Quality of Service (QoS) for carrier gateway applications to guarantee zero packet loss for paid services like voice and video.

The networking subsystem performs standard routing/bridging within the WAN/LAN Ethernet ports at a peak rate of 37.5 Mpps. Advanced features including tunneling and de/fragmentation are performed by a networking processing unit (NPU) that consists of dual 12 threaded programmable engines (UBI32 cores), each running at 1.7 GHz for up to 2.2 Mpps throughput.

IPQ8078 contains an in line security engine with AES 128/256, SHA1-96, 128, 256, and 512 and 3DES for up to 5 Gbps throughput.

1.4 CPU subsystem

The CPU subsystem consists of quad ARM Cortex A53s @ 2.2 GHz, with 64 bit ISA v8 instruction set. The I$/D$ sizes of core are 32kB, while the L2$ is 512kB. Each A53 core has a 64-bit Floating Point/NEON DSP extension that could be used for enhanced audio/voice/video processing.

1.5 Interfaces and power management

IPQ8078 comes with a large variety of interfaces to enable various platform configurations. It has dual PCIe gen2, dual USB3.0, multiple serial IOs selectable between SPI/I2C/UART, Dual SDIO for eMMC and SD card, I2S/PCM/Display Interfaces 16/32 bits DDR3L/4 up to 2400 MT/s, parallel NAND, serial NOR, and Wi-Fi/IOT coexistence interfaces for up to 4 radios.

IPQ8078 comes with advanced power management for lowest active and standby power consumption, making it extremely valuable for carrier gateway and Enterprise AP power over Ethernet (PoE) applications. A companion PMIC PMP8074 is used to optimally manage active/standby power.

Qualcomm 2018-02-01 09:07:11 steve.rumsby@sierrawireless.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

1.6 IPQ8078 features

1.6.1 Wi-Fi subsystem

- 5 GHz antenna configuration
  - 8x8/8s-80MHz or 4x4/4s-80+80MHz
- 2.4 GHz antenna configuration
  - 4x4/4s-40MHz
- Twelve IQ transmit pairs and twelve IQ receive pairs to external QCN51xx
- 802.11ac mode
  - PHY rate: 3466 Mbps (5 GHz) and 800 Mbps (2.4 GHz)
  - 5 GHz: SU-MIMO (8ss, 1 user) and MU-MIMO (8ss, 4 users)
  - 2.4 GHz: SU-MIMO (4ss, 1 user) and MU-MIMO (4ss, 4 users)
  - Explicit beamforming
  - 3.2 µs Symbol Duration; 0.4 µs and 0.8 µs GI
- 802.11ax mode
  - PHY rate: 4800 Mbps (5 GHz) and 1150 Mbps (2.4 GHz)
  - 5 GHz: SU-MIMO (8ss, 1 user), DL MU-MIMO (8ss, 8 user), DL-OFDMA (8 users)
  - 2.4 GHz: SU-MIMO (4ss, 1 user), DL MU-MIMO (4ss, 4 user), DL-OFDMA (8 users)
  - Explicit beamforming
  - 12.8 µs Symbol Duration; 0.8 µs, 1.6 µs, or 3.2 µs GI
- Legacy 11a/b/g/n
- Radio Control interfaces, including Smart Antenna interface to manage external antenna switch
- 4.9 GHz Operation for public safety

1.6.2 Networking subsystem

- 3 SerDes for external Ethernet PHYs
  - Dual up to 10.3125G Ethernet SerDes ports for external 10/5/2.5/1GbE PHYs. Each SerDes can operate in XFI, USXGMII, SGMII+ or SGMII mode
  - Single up to 6.25G Ethernet SerDes for external 5 or 4 ports GbE PHY array or single GbE PHY
- Packet Acceleration
  - Packet Processing Engine (PPE) for standard 5-tuple routing/bridging of IPv4 and IPv6 packets with ingress capacity of 37.5M packet per second (Mpps) and egress capacity of up to 10 Mpps per port

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

- Flexible VLAN assignment and translation on ingress, including filtering, double tag, single tag, untag, priority tag
- Classification based on L2/L3/L4 and User Defined fields; actions like policing, QoS Marking, en-queue, forwarding, and so forth
- Flow based routing/bridging/NAT; IPv4 unicast routing and NAT, IPv6 unicast routing, PPPoE IPMC bridging
- MAC table for Bridge learning and aging, Station Movement control, L2 multicast, Spanning tree, Link aggregation, Egress VLAN filtering, PPPoE
- Egress Queues:
  - 256 unicast and 44 multicast queues
  - PCP, DSCP, Flow, Classifier based Priority
  - Classifier based policer with two rate, three color meter, marker
  - Ingress scheduling, shaping
  - WRED lite with color aware dynamic, and static threshold
  - 2-level Scheduler, 3-level Shaper with CIR, EIR rate control (HTB lite)

- Dual Core Twelve-Threaded network processing unit (NPU) Ubi32 @ 1.7 GHz for up to 2.2 Mpps throughput.
- Wi-Fi driver offload on NPU (optional)
- Up to 64k flows between PPE/NPU/CPU
- 4 level QoS between pipelines

- Security
  - In line security engine
    - Up to 5 Gbps
    - AES 128, 256
    - SHA 1-96, 128, 256, 512
    - 3DES 1-96, MD5-96
    - CCM and GCM operation
    - FIPS level 2 certification
  - 4 OTP keys for multi root revocation
  - SDIO in line crypto
  - Secure execution environment
  - ARM Trustzone

1.6.3 CPU subsystem

- Quad ARM Cortex A53 at 2.2 GHz, 64bits ISA v8 instruction set, 18.4k DMIPS
- 32kB/32kB I$/D$ and 512kB L2$

QUALCOMM® 2015-12-21 06:07:08 PST steve.rumsby@qti.qualcomm.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

- Floating Point & NEON SIMD DSP for each core
- Supports crypto instruction extensions

1.6.4 Peripherals/interfaces

- Dual PCIe-gen2
- Dual USB3.0
- Multiple programmable serial interface for SPI, UART or I2C
- I2S, PCM, and TDMA
- Parallel NAND, eMMC and Display Interface
- Serial NOR
- SD-card
- 16 or 32 bits DDR3L at 1866 MT/s or DDR4 at 2133 MT/s

1.6.5 Power management

- Advanced Power Management for lowest active and stand-by power consumption
- Interface to external PMIC (PMP8074)

1.6.6 Platform extension options

- BT/BLE/15.4 companion chip through SPI/UART
- LTE-WAN through PCIe (or USB)
- 802.11ad through PCIe
- Wi-Fi radio through PCIe
- SLIC through PCM
- Audio Tx/Rx through I2S/TDMA
- Display through QPIC port
- DECT through PCM/SPI
- Storage through USB3.0

1.6.7 Package

- 21 mm × 21 mm 772-pin FCBGA package

1.7 Terms and abbreviations

Table 1-1 lists terms, abbreviations, and acronyms commonly used throughout this document.

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 1-1 Terms and abbreviations

Term	Definition
AP	Access point
DBDC	Dual band dual concurrent
DDR	Double data rate
NPU	Networking processing unit
PPE	Packet processing engine
PoE	Power over Ethernet
PSGMII	Penta-SGMII
QoS	Quality of Service
QPIC	Qualcomm parallel interface controller (NAND + LCD)
QSGMII	Quad-SGMII
SA	Spectrum analysis
SoC	System on a chip
SON	Self-organized network
USXGMII	Universal serial 10G MII
XFI	10G small form factor pluggable interface

QUALCOMM®
2018-02-[illegible] 09:07:08 PST
steve.rumsby@arrent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2 Pin definitions

2.1 I/O parameter definitions

Table 2-1 I/O description parameters

Symbol	Description
Pad attribute	
AI	Analog input (does not include pad circuitry)
AO	Analog output (does not include pad circuitry)
B	Bidirectional digital with CMOS input
DI	Digital input (CMOS)
DO	Digital output (CMOS)
P	Power
H	High-voltage tolerant
S	Schmitt trigger input
Z	High-impedance (high-Z) output
Pad pull details for digital I/Os	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters, and is a prefix to other programmable options: PU:nppdkp = default pull-up, with programmable options following the colon (:). PD:nppukp = default pull-down, with programmable options following the colon (:). NP:pdpukp = default no-pull, with programmable options following the colon (:). KP:nppdpu = default keeper, with programmable options following the colon (:).
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device
NP	Contains no internal pull
KP	Contains an internal week keeper device (keepers cannot drive external buses)
Pad-voltage groupings	
P1	Pad group 1 (EBI/DDR); tied to VDDPX_1 (1.2 V or 1.35 V)
P2	Pad group 2 (SDC2); tied to VDDPX_2 (1.8 V or 2.95 V)
P3	Pad group 3 (general power: mode, JTAG, GPIOS); tied to VDDPX_3 (1.8 V only)
P4	Pad group 4 (2G WSI); tied to VDDPX_4 (1.8 V only)
P7	Pad group 7 (SDC1); tied to VDDPX_7 (1.8 V only)
P8	Pad group 8 (5G WSI); tied to VDDPX_8 (1.8 V only)

QUALCOMM®
2015-02-01 09:07:08 PST
steve.runnerby@spirent.com

Table 2-1 I/O description parameters

Symbol	Description
Output-current drive strength	
EBI pads	Pads for EBI are tailored for 1.2 V interfaces and are source terminated.
3.0 V (H) pads	Programmable drive strength, 2–8 mA, in 2 mA steps
Others[1]	Programmable drive strength, 2–16 mA, in 2 mA steps

1. Digital pads other than EBI0 pads or high-voltage tolerant pads.

2.2 Pin map

The IPQ8078 device is available in the 772-pin FCBGA that includes several ground pins for electrical grounding, mechanical strength, and thermal continuity. See Chapter 4 for package details. A high-level view of the pin assignments is shown in Figure 2-1.

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

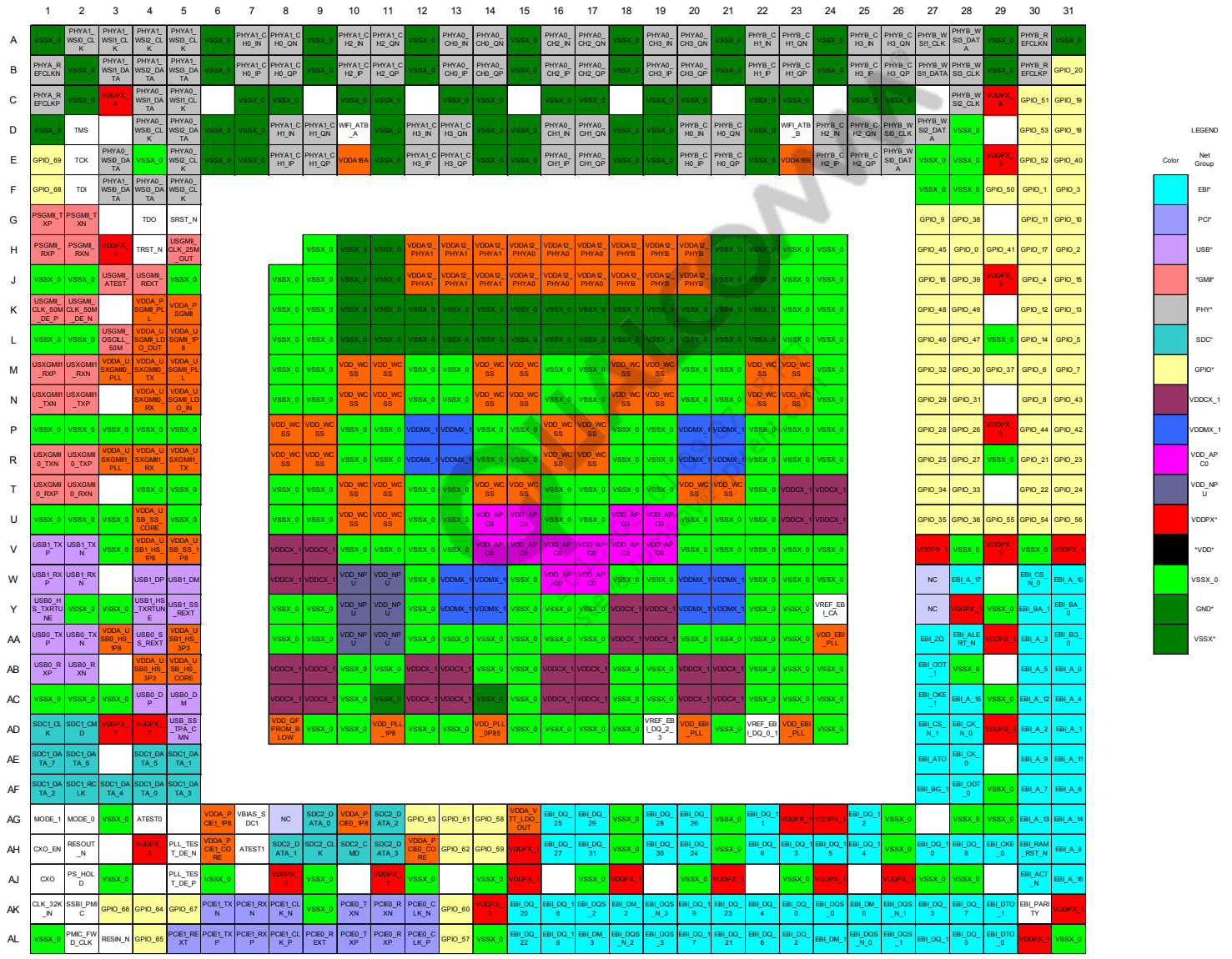


Figure 2-1 IPQ8078 pin assignments

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2.3 Pin descriptions

2.3.1 CLK/RST and PMIC interface

Table 2-2 CLK/RST and PMIC interface

Pad #	Pad name	Voltage	Type	Description
AK1	CLK_32K_IN	1.8 V	DI	Digital sleep clock at 32 KHz
AJ1	CXO	1.8 V	DI	CXO clock input at 19.2 MHz
AH1	CXO_EN	1.8 V	DO	CXO clock enable
AL2	PMIC_FWD_CLK	1.8 V	DO	PMIC clock
AJ2	PS_HOLD	1.8 V	DO	PMIC clock hold
AK2	SSBI_PMIC	1.8 V	B	PMIC Data
AL3	RESIN_N	1.8 V	DI	Hardware reset input
AH2	RESOUT_N	1.8 V	DO	Reset output when chip is in reset

2.3.2 PCI express endpoint

Table 2-3 PCI express endpoint

Pad #	Pad name	Voltage	Type	Description
AK12	PCIE0_CLK_N	0.925 V	AO	Clock to PCIe0 end point.
AL12	PCIE0_CLK_P	0.925 V	AO	
AL9	PCIE0_REXT	–	AI, AO	PCIe0 external reference resistor 100 ohms 1%
AK11	PCIE0_RXN	0.925 V	AI	PCIe0 receive lane
AL11	PCIE0_RXP	0.925 V	AI	
AK10	PCIE0_TXN	0.925 V	AO	PCIe0 transmit lane
AL10	PCIE0_TXP	0.925 V	AO	
AK8	PCIE1_CLK_N	0.925 V	AO	Clock to PCIe1 end point.
AL8	PCIE1_CLK_P	0.925 V	AO	
AL5	PCIE1_REXT	–	AI, AO	PCIe1 external reference resistor 100 ohms 1%
AK7	PCIE1_RXN	0.925 V	AI	PCIe1 receive lane
AL7	PCIE1_RXP	0.925 V	AI	
AK6	PCIE1_TXN	0.925 V	AO	PCIe1 transmit lane
AL6	PCIE1_TXP	0.925 V	AO	

QUALCOMM
steve.[illegible]@spirent.com

2.3.3 Wi-Fi 5 GHz PHY

Table 2-4 Wi-Fi 5 GHz PHY

Pad #	Pad name	Voltage	Type	Description
B1	PHYA_REFCLKN	1.8 V	AI	5G reference clock input
C1	PHYA_REFCLKP	1.8 V	AI	
A13	PHYA0_CH0_IN	1.2 V	AI, AO	5G analog I/Q channel
B13	PHYA0_CH0_IP	1.2 V	AI, AO	
A14	PHYA0_CH0_QN	1.2 V	AI, AO	
B14	PHYA0_CH0_QP	1.2 V	AI, AO	
D16	PHYA0_CH1_IN	1.2 V	AI, AO	
E16	PHYA0_CH1_IP	1.2 V	AI, AO	
D17	PHYA0_CH1_QN	1.2 V	AI, AO	
E17	PHYA0_CH1_QP	1.2 V	AI, AO	
A16	PHYA0_CH2_IN	1.2 V	AI, AO	
B16	PHYA0_CH2_IP	1.2 V	AI, AO	
A17	PHYA0_CH2_QN	1.2 V	AI, AO	
B17	PHYA0_CH2_QP	1.2 V	AI, AO	
A19	PHYA0_CH3_IN	1.2 V	AI, AO	
B19	PHYA0_CH3_IP	1.2 V	AI, AO	
A20	PHYA0_CH3_QN	1.2 V	AI, AO	
B20	PHYA0_CH3_QP	1.2 V	AI, AO	
D4	PHYA0_WSI0_CLK	1.8 V	DO	WSI interface
E3	PHYA0_WSI0_DATA	1.8 V	B	
C5	PHYA0_WSI1_CLK	1.8 V	DO	
C4	PHYA0_WSI1_DATA	1.8 V	B	
E5	PHYA0_WSI2_CLK	1.8 V	DO	
D5	PHYA0_WSI2_DATA	1.8 V	B	
F5	PHYA0_WSI3_CLK	1.8 V	DO	
F4	PHYA0_WSI3_DATA	1.8 V	B	

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-4 Wi-Fi 5 GHz PHY (cont.)

A7	PHYA1_CH0_IN	1.2 V	AI, AO	5G analog I/Q channel
B7	PHYA1_CH0_IP	1.2 V	AI, AO	
A8	PHYA1_CH0_QN	1.2 V	AI, AO	
B8	PHYA1_CH0_QP	1.2 V	AI, AO	
D8	PHYA1_CH1_IN	1.2 V	AI, AO	
E8	PHYA1_CH1_IP	1.2 V	AI, AO	
D9	PHYA1_CH1_QN	1.2 V	AI, AO	
E9	PHYA1_CH1_QP	1.2 V	AI, AO	
A10	PHYA1_CH2_IN	1.2 V	AI, AO	
B10	PHYA1_CH2_IP	1.2 V	AI, AO	
A11	PHYA1_CH2_QN	1.2 V	AI, AO	
B11	PHYA1_CH2_QP	1.2 V	AI, AO	
D12	PHYA1_CH3_IN	1.2 V	AI, AO	
E12	PHYA1_CH3_IP	1.2 V	AI, AO	
D13	PHYA1_CH3_QN	1.2 V	AI, AO	
E13	PHYA1_CH3_QP	1.2 V	AI, AO	
A2	PHYA1_WSI0_CLK	1.8 V	DO	WSI interface
F3	PHYA1_WSI0_DATA	1.8 V	B	
A3	PHYA1_WSI1_CLK	1.8 V	DO	
B3	PHYA1_WSI1_DATA	1.8 V	B	
A4	PHYA1_WSI2_CLK	1.8 V	DO	
B4	PHYA1_WSI2_DATA	1.8 V	B	
A5	PHYA1_WSI3_CLK	1.8 V	DO	
B5	PHYA1_WSI3_DATA	1.8 V	B	
D10	WIFI_ATB_A	1.8 V	AI, AO	5 GHz analog test pin

QUALCOMM®

2.3.4 Wi-Fi 2.4 GHz PHY

Table 2-5 Wi-Fi 2.4 GHz PHY

Pad #	Pad name	Voltage	Type	Description
D20	PHYB_CH0_IN	1.2 V	AI, AO	2.4 GHz analog I/Q channel
E20	PHYB_CH0_IP	1.2 V	AI, AO	
D21	PHYB_CH0_QN	1.2 V	AI, AO	
E21	PHYB_CH0_QP	1.2 V	AI, AO	
A22	PHYB_CH1_IN	1.2 V	AI, AO	
B22	PHYB_CH1_IP	1.2 V	AI, AO	
A23	PHYB_CH1_QN	1.2 V	AI, AO	
B23	PHYB_CH1_QP	1.2 V	AI, AO	
D24	PHYB_CH2_IN	1.2 V	AI, AO	
E24	PHYB_CH2_IP	1.2 V	AI, AO	
D25	PHYB_CH2_QN	1.2 V	AI, AO	
E25	PHYB_CH2_QP	1.2 V	AI, AO	
A25	PHYB_CH3_IN	1.2 V	AI, AO	
B25	PHYB_CH3_IP	1.2 V	AI, AO	
A26	PHYB_CH3_QN	1.2 V	AI, AO	
B26	PHYB_CH3_QP	1.2 V	AI, AO	
A30	PHYB_REFCLKN	1.8 V	AI	2.4 GHz reference clock input
B30	PHYB_REFCLKP	1.8 V	AI	
D26	PHYB_WSI0_CLK	1.8 V	DO	WSI interface
E26	PHYB_WSI0_DATA	1.8 V	B	
A27	PHYB_WSI1_CLK	1.8 V	DO	
B27	PHYB_WSI1_DATA	1.8 V	B	
C28	PHYB_WSI2_CLK	1.8 V	DO	
D27	PHYB_WSI2_DATA	1.8 V	B	
B28	PHYB_WSI3_CLK	1.8 V	DO	
A28	PHYB_WSI3_DATA	1.8 V	B	
D23	WIFI_ATB_B	1.8 V	AI, AO	2.4 GHz analog test pin

QUALCOMM®

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2.3.5 Analog test

Table 2-6 Analog test

Pad #	Pad name	Voltage	Type	Description
AG4	ATEST0	1.8 V	AO	Analog test pin
AH7	ATEST1	1.8 V	AO	

2.3.6 DDR4/DDR3L

Table 2-7 DDR4/DDR3L

Pad #	Pad name	Voltage	Type	Description
AB31	EBI_A_0	1.20/1.35 V	DO	DDR command/address[0:17]
AD31	EBI_A_1	1.20/1.35 V	DO	
W31	EBI_A_10	1.20/1.35 V	DO	
AE31	EBI_A_11	1.20/1.35 V	DO	
AC30	EBI_A_12	1.20/1.35 V	DO	
AG30	EBI_A_13	1.20/1.35 V	DO	
AG31	EBI_A_14	1.20/1.35 V	DO	
AC28	EBI_A_15	1.20/1.35 V	DO	
AJ31	EBI_A_16	1.20/1.35 V	DO	
W28	EBI_A_17	1.20/1.35 V	DO	
AD30	EBI_A_2	1.20/1.35 V	DO	
AA30	EBI_A_3	1.20/1.35 V	DO	
AC31	EBI_A_4	1.20/1.35 V	DO	
AB30	EBI_A_5	1.20/1.35 V	DO	
AF31	EBI_A_6	1.20/1.35 V	DO	
AF30	EBI_A_7	1.20/1.35 V	DO	
AH31	EBI_A_8	1.20/1.35 V	DO	
AE30	EBI_A_9	1.20/1.35 V	DO	
AJ30	EBI_ACT_N	1.20/1.35 V	DO	Activate output
AA28	EBI_ALERT_N	1.20/1.35 V	DI	Alert input
AE27	EBI_ATO	1.20/1.35 V	AO	Analog test output
Y31	EBI_BA_0	1.20/1.35 V	DO	Bank address[0:1]
Y30	EBI_BA_1	1.20/1.35 V	DO	
AA31	EBI_BG_0	1.20/1.35 V	DO	Bank group address[0:1]
AF27	EBI_BG_1	1.20/1.35 V	DO	

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-7 DDR4/DDR3L (cont.)

AE28	EBI_CK_0	1.20/1.35 V	DO	Differential clock
AD28	EBI_CK_N_0	1.20/1.35 V	DO	
AH29	EBI_CKE_0	1.20/1.35 V	DO	Clock enable
AC27	EBI_CKE_1	1.20/1.35 V	DO	
W30	EBI_CS_N_0	1.20/1.35 V	DO	Chip select
AD27	EBI_CS_N_1	1.20/1.35 V	DO	
AK25	EBI_DM_0	1.20/1.35 V	DO	Data mask
AL24	EBI_DM_1	1.20/1.35 V	DO	
AK18	EBI_DM_2	1.20/1.35 V	DO	
AL17	EBI_DM_3	1.20/1.35 V	DO	

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

Table 2-7 DDR4/DDR3L (cont.)

AK23	EBI_DQ_0	1.20/1.35 V	B	Data[0:31]
AL27	EBI_DQ_1	1.20/1.35 V	B	
AH27	EBI_DQ_10	1.20/1.35 V	B	
AG22	EBI_DQ_11	1.20/1.35 V	B	
AG25	EBI_DQ_12	1.20/1.35 V	B	
AH23	EBI_DQ_13	1.20/1.35 V	B	
AH25	EBI_DQ_14	1.20/1.35 V	B	
AH24	EBI_DQ_15	1.20/1.35 V	B	
AK16	EBI_DQ_16	1.20/1.35 V	B	
AL20	EBI_DQ_17	1.20/1.35 V	B	
AL16	EBI_DQ_18	1.20/1.35 V	B	
AK20	EBI_DQ_19	1.20/1.35 V	B	
AL23	EBI_DQ_2	1.20/1.35 V	B	
AK15	EBI_DQ_20	1.20/1.35 V	B	
AL21	EBI_DQ_21	1.20/1.35 V	B	
AL15	EBI_DQ_22	1.20/1.35 V	B	
AK21	EBI_DQ_23	1.20/1.35 V	B	
AH20	EBI_DQ_24	1.20/1.35 V	B	
AG16	EBI_DQ_25	1.20/1.35 V	B	
AG20	EBI_DQ_26	1.20/1.35 V	B	
AH16	EBI_DQ_27	1.20/1.35 V	B	
AG19	EBI_DQ_28	1.20/1.35 V	B	
AG17	EBI_DQ_29	1.20/1.35 V	B	
AK27	EBI_DQ_3	1.20/1.35 V	B	
AH19	EBI_DQ_30	1.20/1.35 V	B	
AH17	EBI_DQ_31	1.20/1.35 V	B	
AK22	EBI_DQ_4	1.20/1.35 V	B	
AL28	EBI_DQ_5	1.20/1.35 V	B	
AL22	EBI_DQ_6	1.20/1.35 V	B	
AK28	EBI_DQ_7	1.20/1.35 V	B	
AH28	EBI_DQ_8	1.20/1.35 V	B	
AH22	EBI_DQ_9	1.20/1.35 V	B	

QUALCOMM®

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-7 DDR4/DDR3L (cont.)

AK24	EBI_DQS_0	1.20/1.35 V	B	Data strobe
AL26	EBI_DQS_1	1.20/1.35 V	B	
AK17	EBI_DQS_2	1.20/1.35 V	B	
AL19	EBI_DQS_3	1.20/1.35 V	B	
AL25	EBI_DQS_N_0	1.20/1.35 V	B	
AK26	EBI_DQS_N_1	1.20/1.35 V	B	
AL18	EBI_DQS_N_2	1.20/1.35 V	B	
AK19	EBI_DQS_N_3	1.20/1.35 V	B	
AL29	EBI_DTO_0	1.20/1.35 V	DO	Digital test output
AK29	EBI_DTO_1	1.20/1.35 V	DO	
AF28	EBI_ODT_0	1.20/1.35 V	DO	On die termination
AB27	EBI_ODT_1	1.20/1.35 V	DO	
AK30	EBI_PARITY	1.20/1.35 V	DO	Parity
AH30	EBI_RAM_RST_N	1.20/1.35 V	DO	Reset output
AA27	EBI_ZQ	–	AI, AO	IO calibration pad (240 ohm 1%)

2.3.7 Mode

Table 2-8 Mode

Pad #	Pad name	Voltage	Type	Description
AG2	MODE_0	1.8 V	DI	Chip functional mode select. Leave unconnected.
AG1	MODE_1	1.8 V	DI	

2.3.8 PLL test clock

Table 2-9 PLL test clock

Pad #	Pad name	Voltage	Type	Description
AH5	PLL_TEST_DE_N	1.8 V	B	PLL higher frequency test pin
AJ5	PLL_TEST_DE_P	1.8 V	B	

QUALCOMM® 2018-02-01 09:07:08 PST steveumsby@spirent.com

2.3.9 PSGMII

Table 2-10 PSGMII

Pad #	Pad name	Voltage	Type	Description
H2	PSGMII_RXN	0.9 V	AI	PSGMII receive data
H1	PSGMII_RXP	0.9 V	AI	
G2	PSGMII_TXN	0.9 V	AO	PSGMII transmit data
G1	PSGMII_TXP	0.9 V	AO	

2.3.10 SDC

Table 2-11 SDC

Pad #	Pad name	Voltage	Type	Description
SDC1				
AD1	SDC1_CLK	1.8 V	DO	Clock
AD2	SDC1_CMD	1.8 V	B	Command
AF4	SDC1_DATA_0	1.8 V	B	Data[0:7]
AE5	SDC1_DATA_1	1.8 V	B	
AF1	SDC1_DATA_2	1.8 V	B	
AF5	SDC1_DATA_3	1.8 V	B	
AF3	SDC1_DATA_4	1.8 V	B	
AE4	SDC1_DATA_5	1.8 V	B	
AE2	SDC1_DATA_6	1.8 V	B	
AE1	SDC1_DATA_7	1.8 V	B	
AF2	SDC1_RCLK	1.8 V	DI	Data strobe for eMMC5.0/5.1 HS400
SDC2				
AH9	SDC2_CLK	1.8/2.95 V	DO	Clock
AH10	SDC2_CMD	1.8/2.95 V	B	Command
AG9	SDC2_DATA_0	1.8/2.95 V	B	Data[0:7]
AH8	SDC2_DATA_1	1.8/2.95 V	B	
AG11	SDC2_DATA_2	1.8/2.95 V	B	
AH11	SDC2_DATA_3	1.8/2.95 V	B	

QUALCOMM®
2018-12-21 09:07:08 PST
stevenlinsby@xplornet.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2.3.11 JTAG

Table 2-12 JTAG

Pad #	Pad name	Voltage	Type	Description
G5	SRST_N	1.8 V	B	JTAG reset for debug
E2	TCK	1.8 V	DI	JTAG test clock input
F2	TDI	1.8 V	DI	JTAG test data input
G4	TDO	1.8 V	DO	JTAG test data output
D2	TMS	1.8 V	DI	JTAG test mode state
H4	TRST_N	1.8 V	DI	JTAG test reset

2.3.12 USB

Table 2-13 USB

Pad #	Pad name	Voltage	Type	Description
USB0				
AD5	USB_SS_TPA_CMN	0.925 V	AI, AO	USB0/1 SS test point
AC5	USB0_DM	3.1 V	AI, AO	USB0 HS data
AC4	USB0_DP	3.1 V	AI, AO	
Y1	USB0_HS_TXRTUNE	–	AI, AO	USB0 HS external resistor 4.02K 1%
AB2	USB0_RXN	0.925 V	AI	USB0 SS receive data
AB1	USB0_RXP	0.925 V	AI	
AA4	USB0_SS_REXT	–	AI	USB0 SS external resistor 100 ohms 1%
AA2	USB0_TXN	0.925 V	AO	USB0 SS transmit data
AA1	USB0_TXP	0.925 V	AO	
USB1				
W5	USB1_DM	3.1 V	AI, AO	USB1 HS data
W4	USB1_DP	3.1 V	AI, AO	
Y4	USB1_HS_TXRTUNE	–	AI, AO	USB1 HS external resistor 4.02K 1%
W2	USB1_RXN	0.925 V	AI	USB1 SS receive data
W1	USB1_RXP	0.925 V	AI	
Y5	USB1_SS_REXT	–	AI	USB1 SS external resistor 100 ohms 1%
V2	USB1_TXN	0.925 V	AO	USB1 SS transmit data
V1	USB1_TXP	0.925 V	AO	

QUALCOMM 2015-02-01 09:05:08 PST stevenumsby@qti.qualcomm.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2.3.13 USXGMII

Table 2-14 USXMII

Pad #	Pad name	Voltage	Type	Description
USXGMII CMN				
J3	USGMII_ATEST	0.9 V	AI, AO	Analog test output
H5	USGMII_CLK_25M_OUT	0.9 V	AO	25MHz clock output
K2	USGMII_CLK_50M_DE_N	0.9 V	AO	50MHz differential clock output
K1	USGMII_CLK_50M_DE_P	0.9 V	AO	
L3	USGMII_OSCILL_50M	0.9 V	AI	Reference clock input
J4	USGMII_REXT	–	AI	USXGMII external resistor 2.49K 1%
USXGMII				
T2	USXGMII0_RXN	0.9 V	AI	USXGMII0 receive data
T1	USXGMII0_RXP	0.9 V	AI	
R1	USXGMII0_TXN	0.85 V	AO	USXGMII0 transmit data
R2	USXGMII0_TXP	0.85 V	AO	
M2	USXGMII1_RXN	0.9 V	AI	USXGMII1 receive data
M1	USXGMII1_RXP	0.9 V	AI	
N1	USXGMII1_TXN	0.85 V	AO	USXGMII1 transmit data
N2	USXGMII1_TXP	0.85 V	AO	

2.3.14 GPIO

Individual GPIO is configured by software using GPIO_CFGn registers corresponding to the GPIO number.

Table 2-15 GPIO pins

Pad #	Pad name	Voltage	Type	GPIO_CFG[5:2] (FUNC_SEL)	Function	Functional description
H28	GPIO_0	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	QPIC_PAD_TE	LCDC TE, VSYNC input
			DO	2	BLSP5_UART_RFR_N	UART5 ready for receiving
			DO	3	BLSP5_I2C_SCL	I2C5 clock
			DO	4	BLSP5_SPI_CLK	SPI5 clock
			DI	5	WCI2_RXD(0)	Wi-Fi WCI 2.0 receive data
			DO	6	BLSP3_SPI_CS1_N(3)	SPI3 additional CS[1]

QUALCOMM 2018-01-01 09:17:08 PST steve.rumsby@silent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

F30	GPIO_1	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	QPIC_PAD_BUSY_N	NAND BUSY_NOT_ READY input. Active low.
			DO	3	MAC1_SA0(2)	Wi-Fi MAC1 smart antenna
			DO	4	BLSP3_SPI_CS1_N(0)	SPI3 additional CS[1]
H31	GPIO_2	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_LCD_RS_N	LCDC RESX, reset signal. Active low.
			DI	2	BLSP5_UART_CTS_N	UART1 clear to send
			B	3	BLSP5_I2C_SDA	I2C5 data
			DO	4	BLSP5_SPI_CS_N	SPI5 chip select
			DO	5	WCI2_TXD(0)	Wi-Fi WCI 2.0 transmit data
			DO	6	BLSP3_SPI_CS2_N(3)	SPI3 additional CS[2]
F31	GPIO_3	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_WE_N	NAND/LCDC write enable
			DO	2	MAC0_SA0(1)	Wi-Fi MAC0 smart antenna
			DO	3	BLSP3_SPI_CS2_N(0)	SPI3 additional CS[2]
J30	GPIO_4	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_OE_N	NAND/LCDC read enable
			DO	2	MAC0_SA1(1)	Wi-Fi MAC0 smart antenna
			DO	3	BLSP3_SPI_CS3_N(0)	SPI3 additional CS[3]
L31	GPIO_5	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(4)	NAND/LCDC data[4]
			DO	2	BLSP4_UART_RFR_ N(0)	UART4 ready for receiving
			DO	3	BLSP4_I2C_SCL(0)	I2C4 clock
			DO	4	BLSP4_SPI_CLK(0)	SPI4 clock
			DO	5	MAC2_SA0(1)	Wi-Fi MAC2 smart antenna

QUALCOMM® 2018-07-01 09:07:08 PDT Steve.Grimsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

M30	GPIO_6	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(5)	NAND/LCDC data[5]
			DI	2	BLSP4_UART_CTS_N(0)	UART4 clear to send
			B	3	BLSP4_I2C_SDA(0)	I2C4 data
			DO	4	BLSP4_SPI_CS_N(0)	SPI4 chip select
			DO	5	MAC2_SA1(1)	Wi-Fi MAC2 smart antenna
M31	GPIO_7	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(6)	NAND/LCDC data[6]
			DI	2	BLSP4_UART_RX(0)	UART4 receive serial data
			DI	3	BLSP4_SPI_MISO(0)	SPI4 master-in slave-out data
N30	GPIO_8	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(7)	NAND/LCDC data[7]
			DO	2	BLSP4_UART_TX(0)	UART4 transmit serial data
			DO	3	BLSP4_SPI_MOSI(0)	SPI4 master-out slave-in data
G27	GPIO_9	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_LCD_CS_N	LCD chip select
			DI	2	BLSP5_UART_RX	UART5 receive serial data
			DI	3	BLSP5_SPI_MISO	SPI5 master-in slave-out data
			DO	4	CXC_CLK(0)	Wi-Fi WSI 1.0 clock
			DO	5	MAC1_SA0(3)	Wi-Fi MAC1 smart antenna
			DO	6	BLSP3_SPI_CS3_N(3)	SPI3 additional CS[3]
G31	GPIO_10	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_CLE_LB_N	NAND CLE/LCDC DCX. CLE is commend latch enable. Active high. DCX is data/commend. 1 is data, 0 is commend.

QUALCOMM 2013-02-01 09:07:08 PST steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

G30	GPIO_11	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_NAND_CS_N	NAND chip select
			DI	2	WCI2_RXD(2)	Wi-Fi WCI 2.0 receive data
			DO	3	MAC1_SA1(2)	Wi-Fi MAC1 smart antenna
K30	GPIO_12	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(1)	NAND/LCDC data[1]
K31	GPIO_13	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(2)	NAND/LCDC data[2]
L30	GPIO_14	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(3)	NAND/LCDC data[3]
J31	GPIO_15	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(0)	NAND/LCDC data[0]
J27	GPIO_16	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_DAT(8)	NAND/LCDC data[8]
			DO	2	BLSP5_UART_TX	UART5 transmit serial data
			DO	4	BLSP5_SPI_MOSI	SPI5 master-out slave-in data
			DO	5	CXC_DATA(0)	Wi-Fi WSI 1.0 data
			DO	6	MAC1_SA1(3)	Wi-Fi MAC1 smart antenna
H30	GPIO_17	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	QPIC_PAD_ALE_LB_N	NAND ALE. Active high.
			DO	2	WCI2_TXD(2)	Wi-Fi WCI 2.0 transmit data
D31	GPIO_18	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	LED(0)	LED interface 0
			DO	2	PWM0(4)	Audio Pulse Width Modulation interface 0
C31	GPIO_19	1.8 V	B	0	GPIO	Configurable I/O
			DOI	1	LED(1)	LED interface 1
			DO	2	PWM1(4)	Audio Pulse Width Modulation interface 1
B31	GPIO_20	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	LED(2)	LED interface 0
			DO	2	PWM2(4)	Audio Pulse Width Modulation interface 2

QUALCOMM® 2015-02-01 09:07:08 PST steve.ramsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

R30	GPIO_21	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PWM0(0)	Audio Pulse Width Modulation interface 0
			DO	2	BLSP4_UART_RFR_N(1)	UART4 ready for receiving
			DO	3	BLSP4_I2C_SCL(1)	I2C4 clock
			DO	4	BLSP4_SPI_CLK(1)	SPI4 clock
			DI	5	WCI2_RXD(3)	Wi-Fi WCI 2.0 receive data
			DO	6	MAC1_SA0(1)	Wi-Fi MAC1 smart antenna
			DO	7	BLSP3_SPI_CS1_N(2)	SPI3 additional CS[1]
T30	GPIO_22	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PWM1(0)	Audio Pulse Width Modulation interface 1
			DI	2	BLSP4_UART_CTS_N(1)	UART4 clear to send
			DO	3	BLSP4_I2C_SDA(1)	I2C4 data
			DO	4	BLSP4_SPI_CS_N(1)	SPI4 chip select
			DO	5	WCI2_TXD(3)	Wi-Fi WCI 2.0 transmit data
			DO	6	MAC1_SA1(1)	Wi-Fi MAC1 smart antenna
			DO	7	BLSP3_SPI_CS2_N(2)	SPI3 additional CS[2]
R31	GPIO_23	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PWM2(0)	Audio Pulse Width Modulation interface 2
			DI	2	BLSP4_UART_RX(1)	UART4 transmit serial data
			DI	3	BLSP4_SPI_MISO(1)	SPI4 master-in slave-out data
			DO	4	BLSP3_SPI_CS3_N(2)	SPI3 additional CS[3]
T31	GPIO_24	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PWM3(0)	Audio Pulse Width Modulation interface 3
			DO	2	BLSP4_UART_TX(1)	UART4 transmit serial data
			DO	3	BLSP4_SPI_MOSI(1)	SPI4 master-out slave-in data

QUALCOMM 2018-07-01 09:07:08 PDT steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

R27	GPIO_25	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	AUDIO_TXMCLK	Master clock source of audio I2S/TDM Tx interface
			DO	2	PWM0(2)	Audio Pulse Width Modulation interface 0
P28	GPIO_26	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	AUDIO_TXBCLK	Bit clock of audio I2S/TDM Tx interface
			DO	2	PWM1(2)	Audio Pulse Width Modulation interface 1
R28	GPIO_27	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	AUDIO_TXFSYNC	Left or right indication of audio I2S Tx interface and frame start indication of Audio TDM Tx interface
			DO	2	PWM2(2)	Audio Pulse Width Modulation interface 2
P27	GPIO_28	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	AUDIO_TXD	Serial digital data output 1 of audio multi-channel I2S Tx interface and serial digital data of audio TDM Tx interface
			DO	2	PWM3(2)	Audio Pulse Width Modulation interface 3
N27	GPIO_29	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	AUDIO_RXMCLK	Master clock source of audio I2S/TDM Rx interface
			DO	2	PWM0(3)	Audio Pulse Width Modulation interface 0
M28	GPIO_30	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	AUDIO_RXBCLK	Bit clock of audio I2S/TDM Rx interface
			DO	2	PWM1(3)	Audio Pulse Width Modulation interface 1

QUALCOMM 2018-02-01 09:07:08 PST steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

N28	GPIO_31	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	AUDIO_RXFSYNC	Left or right indication of audio I2S Rx interface and frame start indication of audio TDM Rx interface
			DO	2	PWM2(3)	Audio Pulse Width Modulation interface 2
M27	GPIO_32	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	AUDIO_RXD	Serial digital data of Audio Rx interface
			DO	2	PWM3(3)	Audio Pulse Width Modulation interface 3
T28	GPIO_33	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PCM_DRX	Transmitted data of audio PCM interface
			DO	2	MAC1_SA0(0)	Wi-Fi MAC1 smart antenna
			DO	3	MAC0_SA0(0)	Wi-Fi MAC0 smart antenna
T27	GPIO_34	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PCM_DTX	Received data of audio PCM interface
			DO	2	MAC1_SA1(0)	Wi-Fi MAC1 smart antenna
			DO	3	MAC0_SA1(0)	Wi-Fi MAC0 smart antenna
U27	GPIO_35	1.8 V	B	0	GPIO	Configurable I/O
			B	1	PCM_FSYNC	Frame start indication of audio PCM interface
			DO	2	MAC2_SA0(0)	Wi-Fi MAC2 smart antenna
U28	GPIO_36	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PCM_PCLK	Clock of audio PCM interface
			DO	2	MAC2_SA1(0)	Wi-Fi MAC2 smart antenna
M29	GPIO_37	1.8 V	B	0	GPIO	Configurable I/O

QUALCOMM® 2018-02-01 09:07:08 PST steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

G28	GPIO_38	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP0_UART_RFR_N	UART0 ready for receiving
			DO	2	BLSP0_I2C_SCL	I2C0 clock
			DO	3	BLSP0_SPI_CLK	SPI0 clock
J28	GPIO_39	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP0_UART_CTS_N	UART0 clear to send
			B	2	BLSP0_I2C_SDA	I2C0 data
			DO	3	BLSP0_SPI_CS_N	SPI3 chip select
E31	GPIO_40	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP0_UART_RX	UART0 receive serial data
			DI	2	BLSP0_SPI_MISO	SPI0 master-in slave-out data
H29	GPIO_41	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP0_UART_TX	UART0 transmit serial data
			DO	2	BLSP0_SPI_MOSI	SPI0 master-out slave-in data
P31	GPIO_42	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP1_UART_RFR_N	UART1 ready for receiving
			DO	2	BLSP1_I2C_SCL	I2C1 clock
			DO	3	BLSP1_SPI_CLK	SPI1 clock
N31	GPIO_43	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP1_UART_CTS_N	UART1 clear to send
			B	2	BLSP1_I2C_SDA	I2C1 data
			DO	3	BLSP1_SPI_CS_N	SPI1 chip select
P30	GPIO_44	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP1_UART_RX	UART1 receive serial data
			DI	2	BLSP1_SPI_MISO	SPI1 master-in slave-out data
H27	GPIO_45	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP1_UART_TX	UART1 transmit serial data
			DO	2	BLSP1_SPI_MOSI	SPI1 master-out slave-in data

QUALCOMM
2018-07-01 09:07:08 PDT
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

L27	GPIO_46	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP2_UART_RFR_N	UART2 ready for receiving
			DO	2	BLSP2_I2C_SCL	I2C2 clock
			DO	3	BLSP2_SPI_CLK	SPI2 clock
L28	GPIO_47	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP2_UART_CTS_N	UART2 clear to send
			B	2	BLSP2_I2C_SDA	I2C2 data
			DO	3	BLSP2_SPI_CS_N	SPI2 chip select
K27	GPIO_48	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP2_UART_RX	UART2 receive serial data
			DI	2	BLSP2_SPI_MISO	SPI2 master-in slave-out data
K28	GPIO_49	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP2_UART_TX	UART2 transmit serial data
			DO	2	BLSP2_SPI_MOSI	SPI2 master-out slave-in data
F29	GPIO_50	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP3_UART_RFR_N	UART3 ready for receiving
			DO	2	BLSP3_I2C_SCL	I2C3 clock
			DO	3	BLSP3_SPI_CLK	SPI3 clock
C30	GPIO_51	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP3_UART_CTS_N	UART3 clear to send
			B	2	BLSP3_I2C_SDA	I2C3 data
			DO	3	BLSP3_SPI_CS_N	SPI3 chip select
E30	GPIO_52	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	BLSP3_UART_RX	UART3 receive serial data
			DI	2	BLSP3_SPI_MISO	SPI3 master-in slave-out data
D30	GPIO_53	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	BLSP3_UART_TX	UART3 transmit serial data
			DO	2	BLSP3_SPI_MOSI	SPI3 master-out slave-in data

QUALCOMM
2018-07-01 09:07:08 PDT
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

U30	GPIO_54	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PTA2_0	Wi-Fi Co-exist PTA 2
			DI	2	WCI2_RXD(1)	Wi-Fi WCI 2.0 receive data
			DO	3	CXC_CLK(1)	Wi-Fi WSI 1.0 clock
			DO	4	BLSP3_SPI_CS1_N(1)	SPI3 additional CS[1]
U29	GPIO_55	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PTA2_1	Wi-Fi Co-exist PTA 2
			DO	2	BLSP3_SPI_CS2_N(1)	SPI3 additional CS[2]
U31	GPIO_56	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PTA2_2	Wi-Fi Co-exist PTA 2
			DO	2	WCI2_TXD(1)	Wi-Fi WCI 2.0 transmit data
			DO	3	CXC_DATA(1)	Wi-Fi WSI 1.0 data
			DO	4	BLSP3_SPI_CS3_N(1)	SPI3 additional CS[3]
AL13	GPIO_57	1.8 V	B	0	GPIO	Configurable I/O
			B	1	PCIE0_CLK_REQ_N	PCIE0 clock request
AG14	GPIO_58	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PCIE0_RST_N	PCIE0 reset
AH14	GPIO_59	1.8 V	B	0	GPIO	Configurable I/O
			B	1	PCIE0_WAKE	PCIE0 wake
AK13	GPIO_60	1.8 V	B	0	GPIO	Configurable I/O
			B	1	PCIE1_CLK_REQ_N	PCIE1 clock request
			DI	2	RX_LOS(2)	Ethernet Loss of Signal
AG13	GPIO_61	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PCIE1_RST_N	PCIE1 reset
AH13	GPIO_62	1.8 V	B	0	GPIO	Configurable I/O
			B	1	PCIE1_WAKE	PCIE1 wake
AG12	GPIO_63	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	SD_CARD_DETECT	SD card detect
			DO	2	PWM0(1)	Audio Pulse Width Modulation interface 0
AK4	GPIO_64	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PTA1_1	Wi-Fi Co-exist PTA 1
			DO	2	PWM1(1)	Audio Pulse Width Modulation interface 1
			DI	4	RX_LOS(1)	Ethernet Loss of Signal

QUALCOMM® 2018-01-01 09:07:08 PST steve.simsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-15 GPIO pins (cont.)

AL4	GPIO_65	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	PTA1_2	Wi-Fi Co-exist PTA 1
AK3	GPIO_66	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	PTA1_0	Wi-Fi Co-exist PTA 1
			DO	2	PWM2(1)	Audio Pulse Width Modulation interface 2
AK5	GPIO_67	1.8 V	B	0	GPIO	Configurable I/O
			DI	1	SD_WRITE_PROTECT	SD card write protection
			DO	2	PWM3(1)	Audio Pulse Width Modulation interface 3
			DI	3	RX_LOS(0)	Ethernet Loss of Signal
F1	GPIO_68	1.8 V	B	0	GPIO	Configurable I/O
			DO	1	MDC	Management Data Clock
E1	GPIO_69	1.8 V	B	0	GPIO	Configurable I/O
			B	1	MDIO	Management Data Input/Output

2.3.15 Ground, power-supply and NC

Table 2-16 Ground, power-supply and NC pins

Pad #	Pad name	Description
Bias, calibration, reference, PLLs power		
AG7	VBIAS_SDC1	Calibration delay circuit
AA24, AD20, AD23	VDD_EBI_PLL	DDR PLL analog power
Y24	VREF_EBI_CA	Reference voltage input for command/address. Provided by resistor divider.
AD22	VREF_EBI_DQ_0_1	Reference voltage input for data lane. Provided by resistor divider.
AD19	VREF_EBI_DQ_2_3	
AD14	VDD_PLL_0P85	PLL analog power 0.85 V
AD11	VDD_PLL_1P8	PLL analog power 1.8 V
AD8	VDD_QFPROM_ BLOW	Power for programming Q-fuses; otherwise connect to ground
Core power		
M10, N11, N14, N15, N18, N19, N22, N23, P8, P9, P16, M11, P17, R8, R9, R16, R17, T10, T11, T14, T15, T20, M14, T21, U10, U11, M15, M18, M19, M22, M23, N10	VDD_WCSS	Wi-Fi core

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-16 Ground, power-supply and NC pins (cont.)

U14, V19, W16, W17, U15, U18, U19, V14, V15, V16, V17, V18	VDD_APC0	Application processor core
AA10, AA11, W10, W11, Y10, Y11	VDD_NPU	NSS Network processor cores
AA18, AB21, AC8, AC9, AC12, AC13, AC16, AC17, AC20, AC21, T23, AA19, T24, U23, U24, V8, V9, W8, W9, Y18, Y19, AB8, AB9, AB12, AB13, AB16, AB17, AB20	VDDCX_1	Digital core
Memory power		
P12, W14, W20, W21, Y13, Y14, Y20, Y21, P13, P20, P21, R12, R13, R20, R21, W13	VDDMX_1	Memory power
PCIE power		
AG10	VDDA_PCIE0_1P8	High voltage supply
AH12	VDDA_PCIE0_ CORE	Analog and digital supply
AG6	VDDA_PCIE1_1P8	High voltage supply
AH6	VDDA_PCIE1_ CORE	Analog and digital supply
USB power		
AB5	VDDA_USB_HS_ CORE	High-speed analog and digital supply
V5	VDDA_USB_SS_1P8	Super-speed high-voltage
U4	VDDA_USB_SS_ CORE	Super-speed analog and digital supply
AA3	VDDA_USB0_HS_ 1P8	High-speed high-voltage
AB4	VDDA_USB0_HS_ 3P3	High-speed high-voltage
V4	VDDA_USB1_HS_ 1P8	High-speed high-voltage
AA5	VDDA_USB1_HS_ 3P3	High-speed high-voltage
USXGMII power		
L5	VDDA_USGMII_1P8	Analog HV power for bandgap and bias generation
N5	VDDA_USGMII_ LDO_IN	LDO input power
L4	VDDA_USGMII_ LDO_OUT	LDO output power
M5	VDDA_USGMII_PLL	Analog power for PLL
M3	VDDA_USXGMII0_ PLL	Analog power for PLL

QUALCOMM 2018-02-01 09:07:08 steve.rumsey@current.com

Table 2-16 Ground, power-supply and NC pins (cont.)

N4	VDDA_USXGMII0_RX	Analog power for Rx
M4	VDDA_USXGMII0_TX	Analog power for Tx
R3	VDDA_USXGMII1_PLL	Analog power for PLL
R4	VDDA_USXGMII1_RX	Analog power for Rx
R5	VDDA_USXGMII1_TX	Analog power for Tx
PSGMII power		
K5	VDDA_PSGMII	Analog power for SGMII
K4	VDDA_PSGMII_PLL	Analog power for PLL
DDR VTT		
AG15	VDDA_VTT_LDO_OUT	VTT LDO output power
5G Wi-Fi power		
H15, H16, H17, J15, J16, J17	VDDA12_PHYA0	Analog 1.2V
H12, H13, H14, J12, J13, J14	VDDA12_PHYA1	Analog 1.2V
E10	VDDA18A	Analog 1.8V
2G Wi-Fi power		
H18, H19, H20, J18, J19, J20	VDDA12_PHYB	Analog 1.2V
E23	VDDA18B	Analog 1.8V
Pad power		
AA29, AJ26, AK31, AL30, V27, V31, Y28, AD29, AG23, AG24, AH15, AJ15, AJ18, AJ21, AJ24	VDDPX_1	Pad power for EBI (DDR)
AJ11	VDDPX_2	Pad power for SDC2
AH4, AJ8, AK14, E29, H3, J29, P29, V29	VDDPX_3	Pad power for General power: MODE, JTAG, GPIOs
C3	VDDPX_4	Pad power for 2G WSI
AD3, AD4	VDDPX_7	Pad power for SDC1
C29	VDDPX_8	Pad power for 5G WSI

QUALCOMM®
2018-02-01 09:37:37 PST
steve.rumsby@sharedocs.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-16 Ground, power-supply and NC pins (cont.)

Ground pad		
A1, A31, B2, B6, B9, B12, B15, B18, B21, B24, B29, A6, C2, C7, C8, C10, C11, C13, C14, C16, C17, C19, A9, C20, C22, C23, C25, C26, D1, D6, D7, D11, D14, A12, D15, D18, D19, D22, E6, E7, E11, E14, E15, E18, A15, E19, E22, H10, H11, H21, H22, J10, J11, J21, J22, A18, K10, K11, K12, K13, K14, K15, K16, K17, K18, K19, A21, K20, K21, K22, L10, L11, L12, L13, L14, L15, L16, A24, L17, L18, L19, L20, L21, L22, A29, AA8, AA21, M17, M20, M21, M24, N8, N9, N12, N13, N16, N17, AA22, N20, N21, N24, P1, P2, P3, P4, P5, P10, P11, AA23, P14, P15, P18, P19, P22, P23, P24, R10, R11, R14, AB10, R15, R18, R19, R22, R23, R24, R29, T4, T5, T8, AB11, T9, T12, T13, T16, T17, T18, T19, T22, U1, U2, AB14, U3, U5, U8, U9, U12, U13, U16, U17, U20, U21, AB15, U22, V3, V10, V11, V12, V13, V20, V21, V22, V23, AB18, V24, V28, V30, W12, W15, W18, W19, W22, W23, W24, AB19, Y2, Y3, Y8, Y9, Y12, Y15, Y16, Y17, Y22, Y23, AB22, Y29, AA9, AB23, AB24, AB28, AC1, AC2, AC3, AC10, AC15, AC18, AC19, AA12, AC22, AC23, AC24, AC29, AD9, AD10, AD12, AD13, AD15, AD16, AA13, AD17, AD18, AD21, AD24, AF29, AG3, AG18, AG21, AG26, AG28, AA14, AG29, AH18, AH21, AH26, AJ3, AJ6, AJ9, AJ12, AJ14, AJ17, AA15, AJ20, AJ23, AJ27, AJ28, AK9, AL1, AL14, AL31, D28, E4, AA16, E27, E28, F27, F28, H9, H23, H24, J1, J2, J5, AA17, J8, J9, J23, J24, K8, K9, K23, K24, L1, L2, AA20, L8, L9, L23, L24, L29, M8, M9, M12, M13, M16, AC14, AC11	VSSX_0	Ground
NC		
W27, Y27, AG8	NC	Not connected pins

2.3.16 Boot configuration GPIOs

Table 2-17 Boot configuration GPIOs

Pad #	Pad name and/or function	Pad name or alt function	Voltage	Type	Description
T31	BOOT_CONFIG[0]	GPIO_24	1.8 V	DI	Auth enable: 0: No auth 1: Auth is required

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-17 Boot configuration GPIOs (cont.)

K28	BOOT_CONFIG[1]	GPIO_49	1.8 V	DI	Fast boot (boot interface select): 0: SPI-NOR 1: eMMC 2: Para NAND 3: USB2.0 4: SPI-NOR-GPT
U28	BOOT_CONFIG[2]	GPIO_36	1.8 V	DI	
U27	BOOT_CONFIG[3]	GPIO_35	1.8 V	DI	
B31	BOOT_CONFIG[4]	GPIO_20	1.8 V	DI	Hash in fuse (SW use only) 0: PK hash is stored in boot ROM. 1: PK hash is stored in OTP.
P27	BOOT_CONFIG[5]	GPIO_28	1.8 V	DI	Boot from ROM: 0: Boot from code RAM 1: Boot from ROM
D30	BOOT_CONFIG[6]	GPIO_53	1.8 V	DI	Boot ROM boot speed: 00: PMIC XO clock (19.2 MHz) 01: GPLL0 (300 MHz) 10: GPLL0 (600 MHz) 11: GPLL0 (800 MHz)
C31	BOOT_CONFIG[7]	GPIO_19	1.8 V	DI	
D31	BOOT_CONFIG[8]	GPIO_18	1.8 V	DI	PK hash source index: 0: From QC efuse 1: From OEM efuse
F1	BOOT_CONFIG[9]	GPIO_68	1.8 V	DI	Use Serial Num: 0: Use Serial Num 1: Use OEM ID
G31	BOOT_CONFIG[10]	GPIO_10	1.8 V	DI	watchdog_enable: 0: Enable watchdog 1: Disable watchdog (default)
H30	BOOT_CONFIG[11]	GPIO_17	1.8 V	DI	DDR type: 0: DDR3 1: DDR4
H27	BOOT_CONFIG[12]	GPIO_45	1.8 V	DI	AARCH64_EN: 0: 32 bits 1: 64 bits (default)
AL4	FORCE_USB_BOOT	GPIO_65	1.8 V	DI	0: Not force to boot from USB 1: Force to boot from USB
Note: All the boot configuration pins are internally weakly pulled down. Pull the pins to the correct values externally on the board.					

QUALCOMM 2018-02-01 09:07:08 PST steve.rumsby@arrivent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

3 Electrical specifications

3.1 Absolute maximum ratings

Absolute maximum ratings (Table 3-1) reflect conditions that the IPQ8078 device may be exposed to beyond the operating limits, without experiencing immediate functional failure. They are limiting values, to be considered individually when all other parameters are within their specified operating ranges. Functionality and long-term reliability can only be expected within the operating conditions, as described in Section 3.2.

Table 3-1 Absolute maximum ratings

Parameter		Min	Max	Unit
Power-supply voltages				
VBIAS_SDC1	Calibration delay circuit	TBD	TBD	V
VDD_EBI_PLL	DDR PLL analog power			
VDD_PLL_0P85	PLL analog power			
VDD_PLL_1P8	PLL analog power			
VDD_QFPROM_BLOW	Power for programming Q-fuses; otherwise connect to ground			
VDDA_PCIE0_1P8	High voltage supply	TBD	TBD	V
VDDA_PCIE0_CORE	Analog and digital supply			
VDDA_PCIE1_1P8	High voltage supply			
VDDA_PCIE1_CORE	Analog and digital supply			
VDDA_USB_HS_CORE	High-speed analog and digital supply	TBD	TBD	V
VDDA_USB_SS_1P8	Super-speed high-voltage			
VDDA_USB_SS_CORE	Super-speed analog and digital supply			
VDDA_USB0_HS_1P8	High-speed high-voltage			
VDDA_USB0_HS_3P3	High-speed high-voltage			
VDDA_USB1_HS_1P8	High-speed high-voltage			
VDDA_USB1_HS_3P3	High-speed high-voltage			
VDDA_USGMII_1P8	Analog HV power for bandgap and bias generation	TBD	TBD	V
VDDA_USGMII_LDO_IN	LDO input power			
VDDA_USGMII_LDO_OUT	LDO output power			
VDDA_USGMII_PLL	Analog power for PLL			
VDDA_USXGMII0_PLL	Analog power for PLL			
VDDA_USXGMII0_RX	Analog power for Rx			
VDDA_USXGMII0_TX	Analog power for Tx			
VDDA_USXGMII1_PLL	Analog power for PLL			
VDDA_USXGMII1_RX	Analog power for Rx			
VDDA_USXGMII1_TX	Analog power for Tx			

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 3-1 Absolute maximum ratings (cont.)

Parameter		Min	Max	Unit
VDDA_PSGMII VDDA_PSGMII_PLL	Analog power for SGMII Analog power for PLL	TBD	TBD	V
VDDA_VTT_LDO_OUT	VTT LDO output power	TBD	TBD	V
VDDA12_PHYA0 VDDA12_PHYA1 VDDA18A	5 GHz analog power 5 GHz analog power 5 GHz analog power	TBD	TBD	V
VDDA12_PHYB VDDA18B	2.4 GHz analog power 2.4 GHz analog power	TBD	TBD	V
VDDPX_1 VDDPX_2 VDDPX_3 VDDPX_4 VDDPX_7 VDDPX_8	Pad power for EBI (DDR) Pad power for SDC2 Pad power for General power: MODE, JTAG, GPIOs Pad power for 2.4 GHz WSI Pad power for SDC1 Pad power for 5 GHz WSI	TBD	TBD	V
VREF_EBI_CA VREF_EBI_DQ_0_1 VREF_EBI_DQ_2_3	Reference voltage input for command/address Reference voltage input for data lane	TBD	TBD	V

3.2 Operating conditions

Operating conditions include design team-controlled parameters such as power supply voltage, power distribution impedances, and thermal conditions (Table 3-2). The IPQ8078 meets all performance specifications listed in Section 3.3 through Section 3.12, when used within the operating conditions, unless otherwise noted in those sections (provided the absolute maximum ratings have never been exceeded).

Table 3-2 Operating conditions

Parameter		Min	Typ[1]	Max	Unit
Power-supply voltages					
VBIAS_SDC1	Calibration delay circuit	1.2	1.25	1.3	V
VDD_EBI_PLL	DDR PLL analog power	1.62	1.8	1.98	V
VDD_PLL_0P85	PLL analog power	0.765	0.85	1.09	V
VDD_PLL_1P8	PLL analog power	1.7	1.8	1.9	V
VDD_QFPROM_BLOW	Power for programming Q-fuses; otherwise connect to ground	1.71	1.8	1.89	V
VDDA_PCIE0_1P8	High voltage supply	1.71	1.8	1.89	V
VDDA_PCIE0_CORE	Analog and digital supply	0.879	0.925	0.971	V
VDDA_PCIE1_1P8	High voltage supply	1.71	1.8	1.89	V
VDDA_PCIE1_CORE	Analog and digital supply	0.879	0.925	0.971	V

QUALCOMM® 2021-02-01 09:07:08 -1 (msby@spirent.com)

Table 3-2 Operating conditions (cont.)

Parameter		Min	Typ[1]	Max	Unit
VDDA_USB_HS_CORE	High-speed analog and digital supply	0.879	0.925	0.971	V
VDDA_USB_SS_1P8	Super-speed high-voltage	1.71	1.8	1.89	V
VDDA_USB_SS_CORE	Super-speed analog and digital supply	0.879	0.925	0.971	V
VDDA_USB0_HS_1P8	High-speed high-voltage	1.71	1.8	1.89	V
VDDA_USB0_HS_3P3	High-speed high-voltage	2.945	3.1	3.255	V
VDDA_USB1_HS_1P8	High-speed high-voltage	1.71	1.8	1.89	V
VDDA_USB1_HS_3P3	High-speed high-voltage	2.945	3.1	3.255	V
VDDA_USGMII_1P8	Analog HV power for bandgap and bias generation	1.71	1.8	1.89	V
VDDA_USGMII_LDO_IN	LDO input power	1.14/1.283	1.2/1.35	1.26/1.417	V
VDDA_USGMII_LDO_OUT	LDO output power	0.81	0.9	0.99	V
VDDA_USGMII_PLL	Analog power for PLL	0.81	0.9	0.99	V
VDDA_USXGMII0_PLL	Analog power for PLL	0.81	0.9	0.99	V
VDDA_USXGMII0_RX	Analog power for Rx	0.765	0.85	0.935	V
VDDA_USXGMII0_TX	Analog power for Tx	0.765	0.85	0.935	V
VDDA_USXGMII1_PLL	Analog power for PLL	0.81	0.9	0.99	V
VDDA_USXGMII1_RX	Analog power for Rx	0.765	0.85	0.935	V
VDDA_USXGMII1_TX	Analog power for Tx	0.765	0.85	0.935	V
VDDA_PSGMII	Analog power for SGMII	0.765	0.85	0.935	V
VDDA_PSGMII_PLL	Analog power for PLL	0.81	0.9	0.99	V
VDDA_VTT_LDO_OUT	VTT LDO output power	0.57/0.642	0.6/0.675	0.63/0.709	V
VDDA12_PHYA0	5 GHz analog power	1.14	1.2	1.26	V
VDDA12_PHYA1	5 GHz analog power	1.14	1.2	1.26	V
VDDA18A	5 GHz analog power	1.71	1.8	1.89	V
VDDA12_PHYB	2.4 GHz analog power	1.14	1.2	1.26	V
VDDA18B	2.4 GHz analog power	1.71	1.8	1.89	V
VDDPX_1	Pad power for EBI (DDR)	1.14/1.283	1.2/1.35	1.26/1.417	V
VDDPX_2	Pad power for SDC2	1.65/2.65	1.8/2.95	1.95/3.1	V
VDDPX_3	Pad power for General power: mode, JTAG, GPIOs	1.65	1.8	1.95	V
VDDPX_4	Pad power for 2.4 GHz WSI	1.65	1.8	1.98	V
VDDPX_7	Pad power for SDC1	1.65	1.8	1.95	V
VDDPX_8	Pad power for 5 GHz WSI	1.65	1.8	1.98	V
VREF_EBI_CA	Reference voltage input for command/address	0.57/0.642	0.6/0.675	0.63/0.709	V
VREF_EBI_DQ_0_1	Reference voltage input for data lane				
VREF_EBI_DQ_2_3					
Thermal conditions					
T_C	Device operating temperature (case)	0	+25	110	°C
	Fuse programming temperature (case)	TBD	TBD	TBD	°C

QUALCOMM® 2018-01-08 07:08 PST Jeevan Kumar@client.com

1. Typical voltages represent the recommended output settings of the companion PMIC device.

Table 3-3 Operating conditions for voltage rails with AVS

Parameter [1]		Min	Typ	Max	Unit
VDD_APC0	Quad Cortex A53 (operating at maximum frequency of 2.0 GHz)				
	Turbo L1	TBD	–	TBD	V
	Nominal	TBD	–	TBD	V
	SVS	TBD	–	TBD	V
VDD_NPU	Turbo	TBD	–	TBD	V
	Nominal	TBD	–	TBD	V
VDD_WCSS	Nominal	TBD	–	TBD	V
VDDCX_1	Nominal	TBD	–	TBD	V
VDDMX_1	Turbo	TBD	–	TBD	V
	Nominal	TBD	–	TBD	V

1. Parts with voltages outside of the specified ranges are not guaranteed to operate properly.

3.3 Power consumption

This information will be included in future revisions of this document.

3.4 Power sequencing

The PMIC includes power-on circuits that provide the proper power sequencing for the entire IPQ8078 chipset. Non-PMIC regulators with power sequence requirement can also be triggered by outputs of the PMIC. For details, refer to the *PMP8074 Power Management Device Specification* (80-P8043-1).

The power-on sequence is shown in Figure 3-1.

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

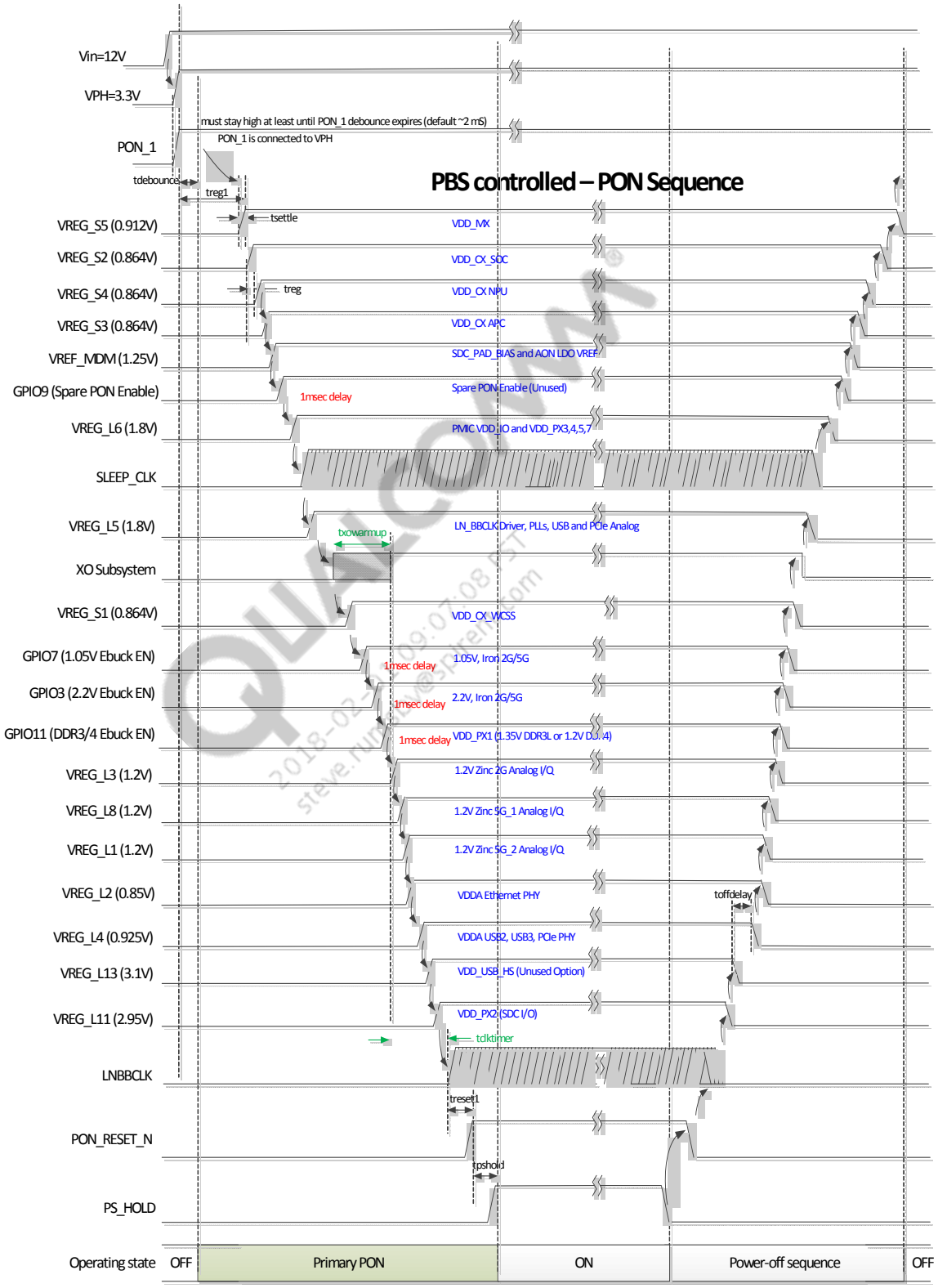


Figure 3-1 Power-on sequence

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

3.5 Digital-logic characteristics

This information will be included in future revisions of this document.

3.6 Timing characteristics

This information will be included in future revisions of this document.

3.7 Memory support

This information will be included in future revisions of this document.

3.8 Connectivity

This information will be included in future revisions of this document.

3.9 UNIPHY interfaces

This information will be included in future revisions of this document.

3.10 Internal functions

Some internal functions require external interfaces to enable their operation. These include clock generation, modes and resets, and JTAG functions.

3.10.1 Clocks

Clocks that are specific to particular functions are addressed in the corresponding sections of this document. Others are specified here.

3.10.1.1 19.2 MHz XO input

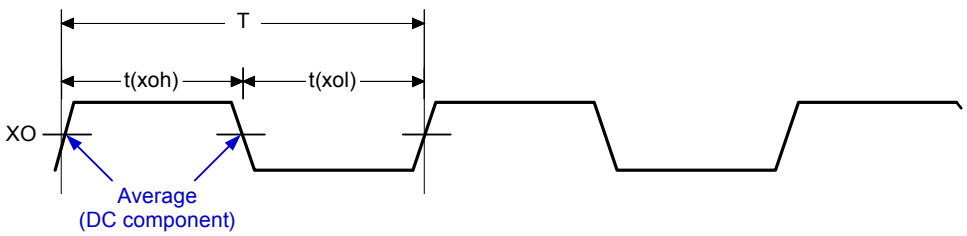


Figure 3-2 XO timing parameters

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 3-4 XO timing parameters

Parameter		Comments [1]	Min	Typ	Max	Unit
t(xoh)	XO logic high		TBD	–	TBD	ns
t(xol)	XO logic low		TBD	–	TBD	ns
T	XO clock period		–	TBD	–	ns
1/T	Frequency	19.2 MHz must be used.	–	TBD	–	MHz

1. Refer the *GPS Quality, 19.2 MHz 2520 Package Size, Crystal, and TH+Xtal Mini-Specification* (80-V9690-24) for more information.

3.10.1.2 Sleep clock

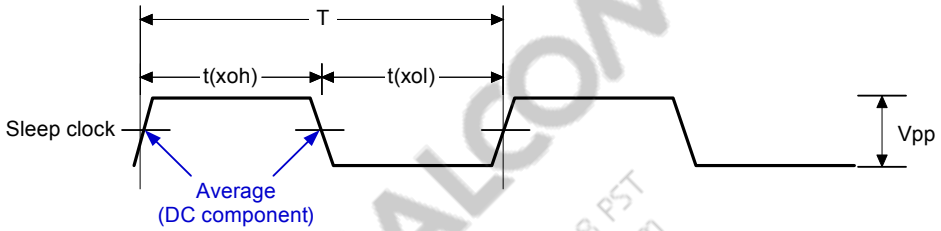


Figure 3-3 Sleep-clock timing parameters

Table 3-5 Sleep-clock timing parameters

Parameter		Comments	Min	Typ	Max	Unit
t(xoh)	Sleep-clock logic high		TBD	–	TBD	µs
t(xol)	Sleep-clock logic low		TBD	–	TBD	µs
T	Sleep-clock period		–	TBD	–	µs
F	Sleep-clock frequency	F = 1/T	–	TBD	–	kHz
Vpp	Peak-to-peak voltage		–	TBD	–	V

3.10.2 Modes and resets

Mode and reset functions are basic digital I/Os that meet the performance specifications presented in Section 3.5.

QUALCOMM® 2018-02-01 09:07:08 PST steve.runge@spirent.com

3.10.3 JTAG

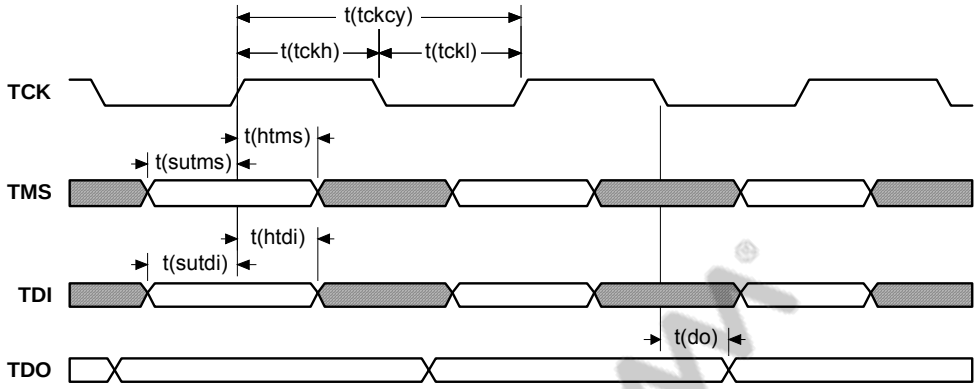


Figure 3-4 JTAG interface timing diagram

Table 3-6 JTAG interface timing characteristics

Parameter		Comments	Min	Typ	Max	Unit
t(tckcy)	TCK period		TBD	–	–	ns
t(tckh)	TCK pulse width high		TBD	–	–	ns
t(tckl)	TCK pulse width low		TBD	–	–	ns
t(sutms)	TMS input setup time		TBD	–	–	ns
t(htms)	TMS input hold time		TBD	–	–	ns
t(sutdi)	TDI input setup time		TBD	–	–	ns
t(htdi)	TDI input hold time		TBD	–	–	ns
t(do)	TDO data output delay		–	–	TBD	ns

3.11 Power management interfaces

3.11.1 SPMI

Table 3-7 Supported SPMI standards and exceptions

Applicable standard	Feature exceptions	MSM variations
MIPI Alliance Specification for System Power Management Interface (SPMI) version 1.0	None	None

3.12 Wi-Fi and Analog interfaces

This information will be included in future revisions of this document.

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

4 Mechanical information

4.1 Device physical dimensions

The IPQ8078 device is available in the 21 mm × 21 mm × 1.8 mm FCBGA package that includes a ground pad for improved grounding, mechanical strength, and thermal continuity. Pin 1 is located by an indicator mark on the top of the package.

Figure 4-1 shows the IPQ8078 device mechanical dimensions, top and bottom views.

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

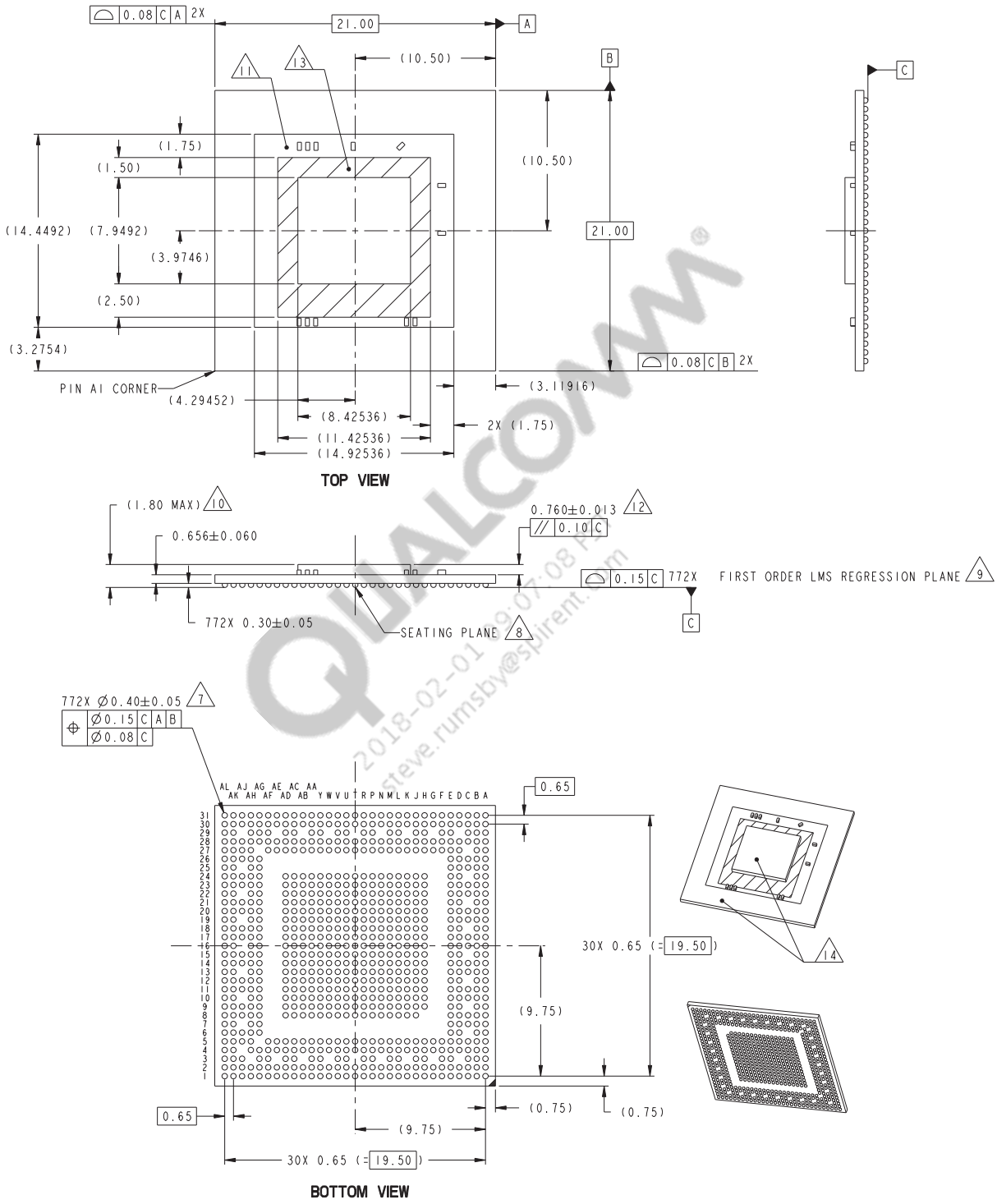


Figure 4-1 IPQ8078 mechanical dimensions, top and bottom views

NOTE Unless otherwise specified:

1. Interpret drawing per ASME Y14.100.

2. All dimensions shown on this drawing are in millimeters (mm).
3. Interpret dimension and tolerances per ASME Y14.5-2009.
4. Workmanship shall be in accordance with Qualcomm package assembly workmanship standard 80-V0691-2.
5. Qualcomm supplied electronic database(s) are for reference only. Dimensional information on current revision of released drawing takes precedence over electronic database(s).
6. Change approval. All changes shall be in accordance with 80-V3652-1 General Supplier Quality Requirement.
7. Dimension measured at the maximum solder ball diameter, parallel to the primary datum -C-.
8. The seating plane is defined by three non-colinear balls that support the free standing package when it is placed on the flat surface. The vectors formed by the three balls establishing the seating plane shall include the center of gravity.
9. Primary datum -C- is determined by the first order LMS regression plane through the spherical crowns of all solder balls on this side of the package.
10. Maximum package height determined by RSS tolerance method.
11. Allowable component area. Maximum component height shall not exceed 0.40 mm.
12. Dimension includes (0.08 mm) bump standoff with underfill.
13. Underfill area, package contact prohibited.
14. Ink not permitted on die and substrate surfaces.

4.2 Part marking

4.2.1 Specification-compliant devices

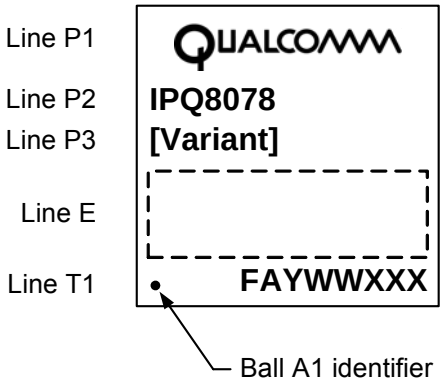


Figure 4-2 IPQ8078 device marking

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 4-1 Package marking line description

Line	Marking	Description
Line P1	QUALCOMM	Qualcomm name
Line P2	IPQ8078	Qualcomm Technologies, Inc. product name
Line P3	[Variant]	Device variant information ■ See Table 4-2 for assigned values.
Line T1	FAYWWXXX	F = source of supply code ■ F = J (Samsung) A = assembly site code ■ A = H (STATSChipPAC, KOREA) ■ A = E (ASE, Taiwan) Y = single/last digit of year WW = two-digit work week of year specified by Y XXX = traceability number
Pin 1	●	Ball A1 identifier
Line E		Space reserved for optional additional trace information

4.3 Device ordering information

Figure 4-3 shows the form of ordering numbers.

Device ID code ►	AAA-AAAA	- P	- CCC	DDDDDD	- EE	- RR	- S	- BB
Symbol definition ►	Product name	Config code	Number of pins	Package type	Shipping package	Product revision	Source code	Feature code
Example ►	IPQ-8078	- 0	- 772	FCBGA	- TR	- 01	- 0	- VV

Feature code (BB) may not be included when identifying older devices.

Figure 4-3 Device identification code

Device identification details for all sample available to date are summarized in Table 4-2.

Table 4-2 Device identification details

Device	Variant (PRR-BB) P = product configuration code RR = product revision code BB = feature code (if applicable)[1]	Shipping package[2]	S value[3]
IPQ8078	001	TR	0
	001	MT	0

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

1. BB is the feature code that identifies an IC's specific feature set, which distinguish it from other versions or variants.
2. TR = tape and reel, MT = matrix tray
3. S is the source configuration code that identifies all the qualified die fabrication-source combinations available at the time a particular sample type was shipped. S values are defined in Table 4-3.

Table 4-3 Source configuration code

S value	Die	F value = J
0	Digital	Samsung

Table 4-4 shows the available ordering numbers.

Table 4-4 Ordering numbers

Ordering number
IPQ-8078-0-772FCBGA-MT-01-0
IPQ-8078-0-772FCBGA-TR-01-0

4.4 Device moisture-sensitivity level

Plastic-encapsulated surface mount packages are susceptible to damage induced by absorbed moisture and high temperature. A package's moisture-sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device.

Qualcomm Technologies Inc. follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. The IPQ8078 is classified as MSL3; the qualification temperature was 255ºC.

4.5 Thermal characteristics

Table 4-5 Device JEDEC thermal resistance

Parameter		Comment	Typ	Unit
θ_{JA}	Junction-to-Ambient	■ With thermal vias ■ JESD51-2A, JESD51-7	18.39	°C/W
θ_{JB}	Junction-to-Board	■ No thermal vias ■ JESD51-7, JESD51-8	9.10	°C/W
θ_{JC}	Junction-to-Case	■ No thermal vias ■ JESD51-7, JESD51-8	0.21	°C/W

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5 Carrier, storage, and handling information

5.1 Carrier

5.1.1 Tape and reel information

Carrier tape system conforms to the EIA-481 standard.

Simplified sketches of the IPQ8078 tape carrier is shown in Figure 5-1 and Figure 5-2, including the part orientation. Tape and reel details for the IPQ8078 are as follows:

- Reel diameter: 330 mm
- Hub size: 178 mm
- Tape width: 32 mm
- Tape pocket pitch: 24 mm
- Feed: Dual
- Units per reel: 500

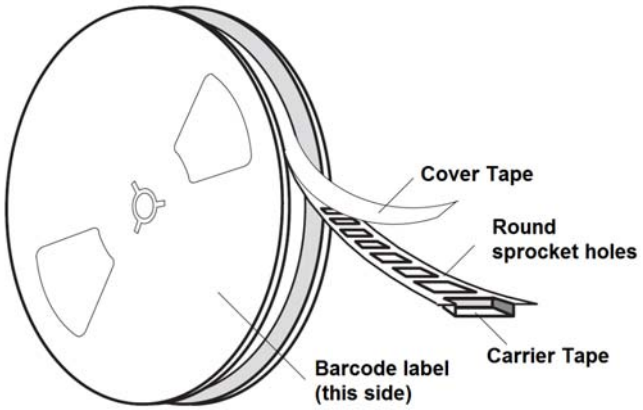


Figure 5-1 Tape orientation on reel

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

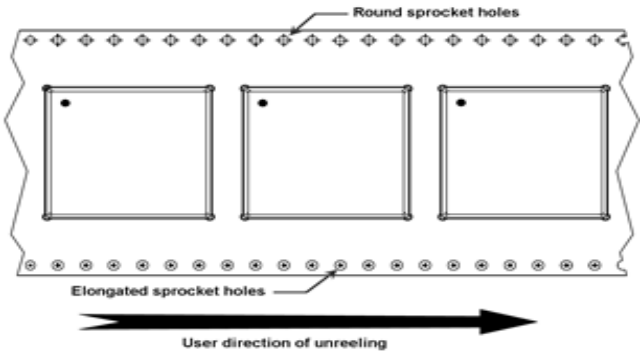
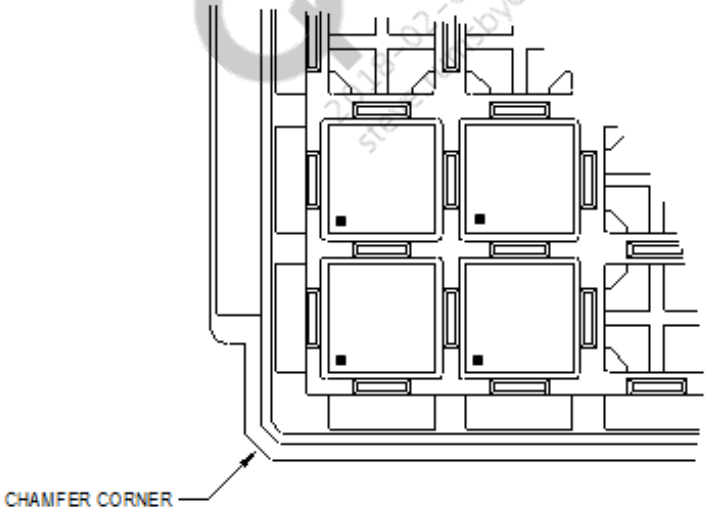


Figure 5-2 Part orientation in tape

5.1.2 Matrix tray information

Qualcomm Technologies matrix tray carriers conform to JEDEC standards. The device pin 1 is oriented to the chamfered corner of the matrix tray. Each tray of the IPQ8078 device contains up to 60 devices. See Figure 5-3 for matrix-tray key attributes and dimensions.



Key dimensions	
Array	5 × 12 = 60
M	16.59 mm
M1	16.59 mm
M2	25.62 mm
M3	25.68 mm

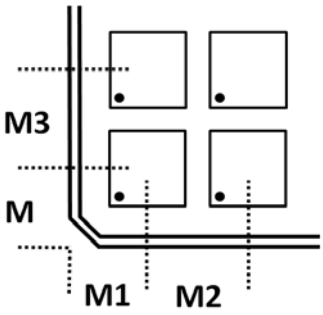


Figure 5-3 Matrix tray part orientation

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5.2 Storage

5.2.1 Bag storage conditions

IPQ8078 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, anti-static bags. Refer to the *ASIC Packing Methods and Materials Specification* (80-VK055-1) for the expected shelf life.

5.2.2 Out of bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in Section 4.4.

5.3 Handling

Tape handling is described in Section 5.1.1. Other (IC-specific) handling guidelines are presented below.

5.3.1 Baking

It is **not necessary** to bake the IPQ8078 if the conditions specified in Section 5.2.1 and Section 5.2.2 have **not been exceeded**.

It is **necessary** to bake the IPQ8078 if any condition specified in Section 5.2.1 or Section 5.2.2 has **been exceeded**. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see *ASIC Packing Methods and Materials Specification* (80-VK055-1) for details.

CAUTION If baking is required, the devices must be transferred into trays that can be baked to at least 125°C. Devices should not be baked in tape and reel carriers at any temperature.

5.3.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

Products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, *Protection of Electrical and Electronic Parts, Assemblies, and Equipment*.

Refer to Section 7.1 for the ESD ratings.

5.4 Barcode label and packing for shipment

Refer to the *ASIC Packing and Materials Specification* (80-VK055-1) document for all packing-related information, including barcode label details.

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

6 PCB mounting guidelines

6.1 RoHS compliance

The IPQ8078 complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC305 composition. A Product Material Declaration (PMD), which provides RoHS and other product environmental governance information, will be published when the data are available.

6.2 SMT parameters

This section describes board-level characterization process parameters. It is included to assist customers with their SMT process development; it is not intended to be a specification for their SMT processes.

6.2.1 Land pad and stencil design

The land pattern and stencil recommendations presented in this section are based on Qualcomm internal characterizations for lead-free solder pastes on an eight-layer PCB built primarily to the specifications described in JEDEC JESD22-B111.

Qualcomm recommends characterizing the land patterns according to each customer's processes, materials, equipment, stencil design, and reflow profile prior to PCB production. Optimizing the solder stencil pattern design and print process is critical to ensure print uniformity, decrease voiding, and increase board-level reliability.

General land pattern guidelines:

- Non-solder-mask-defined (NSMD) pads provide the best reliability.
- Keep the solderable area consistent for each pad, especially when mixing via-in-pad and non-via-in-pad in the same array.
- Avoid large solder mask openings over ground planes.
- Traces for external routing are recommended to be less than or equal to half the pad diameter to ensure consistent solder joint shapes.

One key parameter that should be evaluated is the ratio of aperture area to sidewall area – the area ratio (AR).

For more information, refer to *PCB Land and Stencil Design Guide* (LS90-NG134-1).

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

6.2.2 Reflow profile

Reflow profile conditions typically used by Qualcomm for lead-free systems are listed in Table 6-1 and are shown in Figure 6-1.

Table 6-1 Typical SMT reflow profile conditions (for reference only)

Profile stage	Description	Temp range	Condition
Preheat	Initial ramp	< 150°C	3°C/sec max
Soak	Flux activation	150 to 190°C	60 to 75 sec
Ramp	Transition to liquidus (solder-paste melting point)	190 to 220°C	< 30 sec
Reflow	Time above liquidus	220 to 245°C [1]	50 to 70 sec
Cool down	Cool rate – ramp to ambient	< 220°C	6°C/sec max

1. During the reflow process, the recommended peak temperature is 245°C (minimum). This temperature should not be confused with the peak temperature reached during MSL testing, as described in Section 6.2.3.

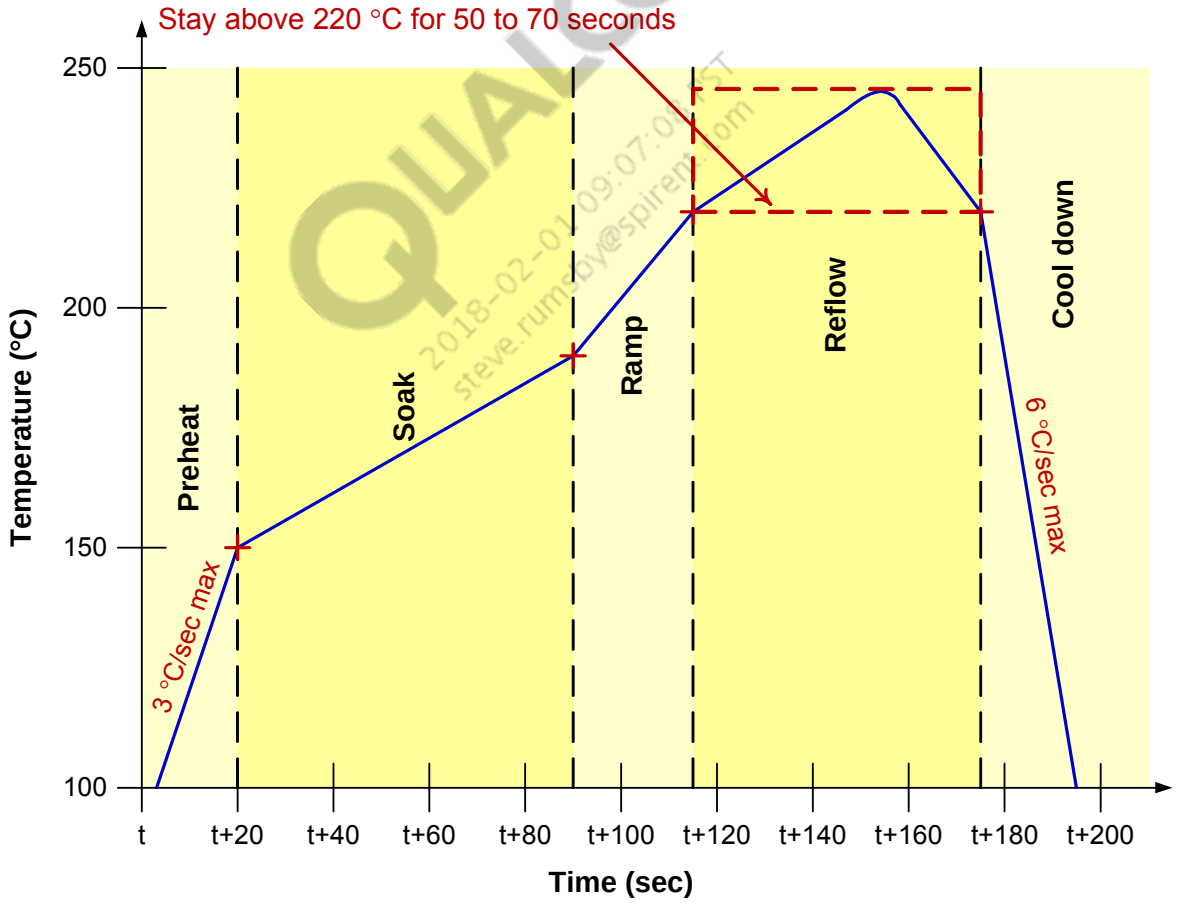


Figure 6-1 Typical SMT reflow profile

6.2.3 SMT peak package-body temperature

This document states a peak package-body temperature in three other places within this document; without explanation, they may appear to conflict. The three places are listed below, along with an explanation of the stated value and its meaning within that section's context.

1. Section 4.4 – *Device moisture-sensitivity level*

   IPQ8078 devices are classified as MSL3@255°C. The temperature (255°C) included in this designation is the lower limit of the range stated for moisture resistance testing during the device qualification process, as explained in #2 below.

2. Section 7.1 – *Reliability qualification summary*

   One of the tests conducted for device qualification is the moisture resistance test. Qualcomm follows J-STD-020-C, and hits a peak reflow temperature that falls within the range of 260°C +0/-5°C (255°C to 260°C).

3. Section 6.2.2 – *Reflow profile*

   During a production board's reflow process, the temperature seen by the package must be controlled. Obviously the temperature must be high enough to melt the solder and provide reliable connections, but it must not go so high that the device might be damaged. The recommended peak temperature during production assembly is 245°C. This is comfortably above the solder melting point (220°C), yet well below the proven temperature reached during qualification (255°C or more).

6.2.4 SMT process verification

Qualcomm recommends verification of the SMT process prior to high-volume board assembly, including:

- In-line solder-paste deposition monitoring
- Reflow-profile measurement and verification
- Visual and x-ray inspection after soldering to confirm adequate alignment, solder voids, solder-ball shape, and solder bridging
- Cross-section inspection of solder joints for wetting, solder-ball shape, and voiding

6.2.5 Board-level reliability

Qualcomm conducts characterization tests to assess the device's board-level reliability, including the following physical tests on evaluation boards:

- Drop shock (JESD22-B111)
- Temperature cycling (JESD22-A104)
- Cyclic bend testing – optional (JESD22-B113)

For board-level reliability data, refer to *Board-Level Reliability 519 FCBGA* (BR80-Y0791-1).

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

6.3 High-temperature warpage

Qualcomm measures high-temperature warpage using a shadow moire system. For detailed data, refer to *High-Temperature Warpage 519 FCBGA* (WR80-Y0791-1).

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

7 Part reliability

7.1 Reliability qualifications summary

This information will be included in future revisions of this document.

7.2 Qualification sample description

- Device name: IPQ8078
- Package type: FCBGA772
- Package body size: 21 mm × 21 mm × 1.8 mm
- Lead count: 772
- Lead pitch: 0.65 mm

QUALCOMM®
2018-02-01 09:07:08 PST
steve.rumsby@spirent.com